About the Project

Introduction

The Electronic Commerce or eCommerce is a term for a business or business exchange, that includes the change of data over the internet. Online business permits clients to electronically trade products and ventures with no boundary of time or distance. The Mobile Commerce or mCommerce is the purchasing and selling of goods and services through wireless handheld gadgets, for example, mobiles and tablets. It is known as next generation eCommerce. Online shopping is a type of eCommerce which permits clients to directly purchase goods or services from a merchant over the internet using a web browser.

Traditional shopping is a tedious and time-consuming job. Although the growing trend of online shopping has reduced some load, there is still some difference in actually going to
shops and hand-picking products to get the feel of their quality and features that cannot be experienced online. Customers also feel worried to carry out online purchases due to fear of less secure transaction process that may lead to hacking of user's sensitive data, insecurity of credit/debit cards, unreliability or breach of privacy. The project aims at removing flaws of both kinds of shopping and bridge the gap between physical and a virtual world.

Pros & Cons of Online Shopping

1) Pros:

- You can do shopping 24/7 any time
- You can sit in the comfort of your home
- Manny stores are within a click away
- Doing the comparison for shopping is easier
- You can get the discount coupons online
- Don't have to drag or keep an eye on kids out in the shop
- No waiting lines on the cash counter
- You get maximum privacy
- You save on fuel for vehicle

2) Cons:

- You can only view items on one page at a time
- It may take the long time with a slow internet connection
- You are not able to touch or feel the products
- You are charged for shipping and handling cost
- Their may be online transaction insecurity
- You may lack the personal supervision of the item

Objective

The main objective of this project is to develop an Android Application "Utterly" .The Android Application will be an Online Shopping Cart Application which will be the eCommerce Shopping. The app is build using the Android Studio software which is the official Google IDE for Android development. It is an open-source software thus increasing cost efficiency, and provides various tools for app development on every android
device possible. The main advantage of this software is the availability of the real-time platform called emulator tool to test our app. Apart from that, the app can also be tested on smartphones through USB debugging and uses SQLite database management system to store the data.

Hardware and Software Requirements

Hardware Requirements

1. Smartphone with Android OS version 6.0 (Marshmallow) or higher
2. Minimum 512 MB of RAM
3. A processor with speeds above 1.2 GHz (any make)
4. 16 MB of storage for the app and extra for the data stored, the size of the app increases as the number of entries are increased
5. Android API version 19
6. Permission to install applications over USB and installation from unknown sources from 'Developer Options'

Software Requirements

Android Application Development is possible with a couple of software and development kits to support the software and execution, they are as follows,

1. ***Android Studio:*** Android studio is the official Integrated Development Environment (IDE) for designing, coding, debugging and executing applications for Google's Smartphone operating system, Android. It has all keywords inbuilt for ease in back-end programming and also design statements, for ease in designing as well. The version of Android Studio used to design the application is 1.4.1.to 3.1 as we are working on studios 3.1.
2. ***Java Development Kit (JDK):*** Since Android applications require Java programming for its backend programming; it needs a JAVA environment to support its functions, executions and syntax.
3. ***Xml:*** The front-end design of the application involves xml statements for the Relative layouts, Radio buttons, Radio Group, buttons, text boxes and text views.

Introduction to Andriod

Android is a mobile operating system based on a modified version of the Linux kernel and other open source software, designed primarily for touchscreen mobile devices such as smartphones and tablets. Android is developed by a consortium of developers known as the Open Handset Alliance and commercially sponsored by Google. It was unveiled in November 2007, with the first commercial Android device, the HTC Dream, being launched in September 2008.

Most versions of Android are proprietary. The core components are taken from the Android Open Source Project (AOSP), which is free and open-source software primarily licensed under the Apache License. When Android is actually installed on devices, ability to modify the otherwise FOSS software is usually restricted, either by not providing the corresponding source code or preventing reinstallation through technical measures, rendering the installed version proprietary. Most Android devices ship with additional proprietary software pre-installed, most notably Google Mobile Services (GMS) which includes core apps such as Google Chrome, the digital distribution platform Google Play, and associated Google Play Services development platform.

Over 70 percent of Android smartphones run Google's ecosystem; some with vendor-customized user interface and software suite, such as *TouchWiz* and later *One UI* by Samsung, and *HTC Sense*. Competing Android ecosystems and forks include Fire OS (developed by Amazon) or LineageOS. However the "Android" name and logo are trademarks of Google which imposes standards to restrict the use of Android branding by "uncertified" devices outside their ecosystem.

History

Android Inc. was founded in Palo Alto, California, in October 2003 by Andy Rubin, Rich Miner, Nick Sears, and Chris White. Rubin described the Android project as having "tremendous potential in developing smarter mobile devices that are more aware of its owner's location and preferences". The early intentions of the company were to develop an advanced operating system for digital cameras, and this was the basis of its pitch to investors in April 2004. The company then decided that the market for cameras was not large enough for its goals, and five months later

it had diverted its efforts and was pitching Android as a handset operating system that would rival Symbian and Microsoft Windows Mobile.

Rubin had difficulty attracting investors early on, and Android was facing eviction from its office space. Steve Perlman, a close friend of Rubin, brought him $10,000 in cash in an envelope, and shortly thereafter wired an undisclosed amount as seed funding. Perlman refused a stake in the company, and has stated "I did it because I believed in the thing, and I wanted to help Andy."

In 2005, Rubin tried to negotiate deals with Samsung and HTC. Shortly afterwards, Google acquired the company in July of that year for at least $50 million; this was Google's "best deal ever" according to Google's then-vice president of corporate development, David Lawee, in 2010. Android's key employees, including Rubin, Miner, Sears, and White, joined Google as part of the acquisition. Not much was known about the secretive Android Inc. at the time, with the company having provided few details other than that it was making software for mobile phones. At Google, the team led by Rubin developed a mobile device platform powered by the Linux kernel. Google marketed the platform to handset makers and carriers on the promise of providing a flexible, upgradeable system. Google had "lined up a series of hardware components and software partners and signaled to carriers that it was open to various degrees of cooperation"

Interface

Android's default user interface is mainly based on direct manipulation, using touch inputs that loosely correspond to real-world actions, like swiping, tapping, pinching, and reverse pinching to manipulate on-screen objects, along with a virtual keyboard. Game controllers and full-size physical keyboards are supported via Bluetooth or USB. The response to user input is designed to be immediate and provides a fluid touch interface, often using the vibration capabilities of the device to provide haptic feedback to the user. Internal hardware, such as accelerometers, gyroscopes and proximity sensors are used by some applications to respond to additional user actions, for example adjusting the screen from portrait to landscape depending on how the device is oriented, or allowing the user to steer a vehicle in a racing game by rotating the device, simulating control of a steering wheel.

Application

Applications ("apps"), which extend the functionality of devices (and must be 64-bit), are written using the Android software development kit and, often, Kotlin programming language, which replaced Java as Google's preferred language for Android app development in May 2019, and was originally announced in May 2017. Java is still supported (originally the only option for user-space programs, and is often mixed with Kotlin), as is C++. Java or other JVM languages, such as Kotlin, may be combined with C/C++, together with a choice of non-default runtimes that allow better C++ support.[116] The Go programming language is also supported, although with a limited set of application programming interfaces (API).

The SDK includes a comprehensive set of development tools,[118] including a debugger, software libraries, a handset emulator based on QEMU, documentation, sample code, and tutorials. Initially, Google's supported integrated development environment (IDE) was Eclipse using the Android Development Tools (ADT) plugin; in December 2014, Google released Android Studio, based on IntelliJ IDEA, as its primary IDE for Android application development. Other development tools are available, including a native development kit (NDK) for applications or extensions in C or C++, Google App Inventor, a visual environment for novice programmers, and various cross platform mobile web applications frameworks. In January 2014, Google unveiled a framework based on Apache Cordova for porting Chrome HTML 5 web applications to Android, wrapped in a native application shell. Additionally, Firebase was acquired by Google in 2014 that provides helpful tools for app and web developers.

Android has a growing selection of third-party applications, which can be acquired by users by downloading and installing the application's APK (Android application package) file, or by downloading them using an application store program that allows users to install, update, and remove applications from their devices. Google Play Store is the primary application store installed on Android devices that comply with Google's compatibility requirements and license the Google Mobile Services software. Google Play Store allows users to browse, download and update applications published by Google and third-party developers; as of January 2021, there are more than three million applications available for Android in Play Store. As of July 2013, 50 billion application installations had been performed. Some carriers offer direct carrier billing

for Google Play application purchases, where the cost of the application is added to the user's monthly bill. As of May 2017, there are over one billion active users a month for Gmail, Android, Chrome, Google Play and Maps.

Introduction of SQLite

SQLite is an open source, embedded relational database. SQLite originally released in 2000, it is designed to provide a suitable way for applications to manage data without the overhead that often comes with dedicated relational database management systems.
In the simplest terms, SQLite is a free, open source, public-domain software package that provides a RDBMS (relational database management system). RDBMS are used to store user-defined records in large tables. In data storage and management system, a database engine can process complex query commands that combine data from multiple tables to generate reports and data summary.

Other popular RDBMS servers include Oracle Database, IBM's DB2, and Microsoft's SQL Server on the commercial uses, with MySQL and PostgreSQL being popular open source products. The SQLite library accesses its storage files directly. Zero Configuration No server means no setup.

To understand the concept better, check out Great Learning Academy's free course on **SQLite for Data Science**. A certification will improve your chances of getting hired.

SQLite History

Hipp and his team creating a simple embedded SQL database in January 2000, that would use for GNU DBM B-Tree library (gdbm) as a back-end, that would not require for any installation or administrative support whatsoever. in August 2000, SQLite 1.0 was released.

SQLite version 2to3 upgrade in 2004. SQLite primary goal was enhanced internationalization support for UTF-8 and UTF-16 text. In 2011 Hipp announced to add UNQI interface to SQLite DB and to develop UNQLite.

Uses SQLite

Today, SQLite is used in a spacious variety of software and products. It is also used in Apple's Mac OS operating system as a part of their Core Data application framework. and also used in the system's Safari web browser, Mail.app email program, RSS manager, as well as Apple's Aperture photography software.

SQLite can be viewed in Sun's Solaris OS (operating system) environment, specifically the database backing the Service Management Facility that debuted with Solaris 10, a core component of its predictive is self-healing technology. SQLite, has been added as a part of PHP 5 standard library.

Features and Limitation of SQLite

SQLite is defined by the following features:

Serverless: –

SQLite does not require a server process or system to operate database. The SQLite library accesses its storage files directly.

Zero Configuration: –

No server means no setup. SQLite is not required to install any application, configure, and nothing to worry about.

Cross-Platform: –

The entire database instance resides in a single cross-platform file, it is not required any administration.

Self-Contained: –

A single library contains the entire database system, which integrated directly into a host application.

Small Runtime Footprint: –

The default build is less than a megabyte of code and requires only a few megabytes of memory. With some adjustments, both the library size and memory use can be significantly reduced.

Transactional: –

SQLite transactions are follows ACID property, it is allowing safe access from multiple processes or threads.

Full-Featured: –

SQLite supports most of the query language features found in the SQL92 (SQL2) standard.

Highly Reliable: –

The purpose of a database is to keep your data safe and organized. To maintain a high level – reliability of database, the core SQLite library is tested before each release. In full, the standard SQLite test suites consist of over 10 million-unit tests and query tests.

Limitations

Foreign key constraints: – Foreign keys are the foundation of referential integrity in relational databases. While SQLite parses them, it currently does not have support for foreign keys. It does support check constraints, and foreign key support is estimated to be completed by sometime in 2006.

Trigger support: –There is some support for triggers but it is not complete. Missing features include FOR EACH STATEMENT triggers (currently all triggers must be FOR EACH ROW), INSTEAD OF triggers on tables (currently INSTEAD OF triggers are only allowed on views), and recursive triggers—triggers that trigger themselves.

ALTER TABLE support: – Only the RENAME TABLE and ADD COLUMN variants of the ALTER TABLE command are supported. Other kinds of ALTER TABLE operations such as DROP COLUMN, ALTER COLUMN, and ADD CONSTRAINT are not implemented.

Nested transactions: – SQLite allows only a single transaction to be active at one time. Nested transactions allow for fine-grained control over larger, more complex operations in that parts of a transaction can be defined and rolled back in case of an error rather than the entire transaction.

RIGHT and FULL OUTER JOIN: – LEFT OUTER JOIN is implemented, but RIGHT OUTER JOIN and FULL OUTER JOIN are not implemented. LEFT OUT JOIN can be implemented as a right outer join by simplified reversing the order of the tables and modify the join constraint. Furthermore, FULL OUTER JOIN can be implemented as a combination of other relational operations supported by SQLite.

Updatable views: – VIEWs in SQLite are read-only. You may not execute a DELETE, INSERT, or UPDATE statement on a view. But you can create a trigger that fires on an attempt to DELETE, INSERT, or UPDATE a view and do what you need in the body of the trigger.

GRANT and REVOKE: – GRANT and REVOKE commands in general are aimed at much higher end systems where there are multiple users who have varying access levels to data in the database.

Introduction to Java

JAVA was developed by James Gosling at **Sun Microsystems**_Inc in the year **1995**, later acquired by Oracle Corporation. It is a simple programming language. Java makes writing, compiling, and debugging programming easy. It helps to create reusable code and modular programs. Java is a class-based, object-oriented programming language and is designed to have as few implementation dependencies as possible. A general-purpose programming language made for developers to *write once run anywhere* that is compiled Java code can run on all platforms that support Java. Java applications are compiled to byte code that can run on any Java Virtual Machine. The syntax of Java is similar to c/c++.

History: Java's history is very interesting. It is a programming language created in 1991. James Gosling, Mike Sheridan, and Patrick Naughton, a team of Sun engineers known as the **Green team** initiated the Java language in 1991. **Sun Microsystems** released its first public implementation in 1996 as **Java 1.0**. It provides no-cost -run-times on popular platforms. Java1.0 compiler was re-written in Java by Arthur Van Hoff to strictly comply with its specifications. With the arrival of Java 2, new versions had multiple configurations built for different types of platforms.

In 1997, Sun Microsystems approached the ISO standards body and later formalized Java, but it soon withdrew from the process. At one time, Sun made most of its Java implementations available without charge, despite their proprietary software status. Sun generated revenue from Java through the selling of licenses for specialized products such as the Java Enterprise System.

On November 13, 2006, Sun released much of its Java virtual machine as free, open-source software. On May 8, 2007, Sun finished the process, making all of its JVM's core code available under open-source distribution terms.

The principles for creating java were simple, robust, secured, high performance, portable, multi-threaded, interpreted, dynamic, etc. In 1995 Java was developed by **James Gosling**, who is known as the Father of Java. Currently, Java is used in mobile devices, internet programming, games, e-business, etc.

Java Terminology

Before learning Java, one must be familiar with these common terms of Java.

1. Java Virtual Machine(JVM): This is generally referred to as JVM. There are three execution phases of a program. They are written, compile and run the program.

- Writing a program is done by a java programmer like you and me.
- The compilation is done by the **JAVAC** compiler which is a primary Java compiler included in the Java development kit (JDK). It takes the Java program as input and generates bytecode as output.
- In the Running phase of a program, **JVM** executes the bytecode generated by the compiler.

Now, we understood that the function of Java Virtual Machine is to execute the bytecode produced by the compiler. Every Operating System has a different JVM but the output they produce after the execution of bytecode is the same across all the operating systems. This is why Java is known as a **platform-independent language.**

2. Bytecode in the **Development process:** As discussed, the Javac compiler of JDK compiles the java source code into bytecode so that it can be executed by JVM. It is saved as **.class** file by the compiler. To view the bytecode, a disassembler like javap can be used.

3. Java Development Kit(JDK): While we were using the term JDK when we learn about bytecode and JVM. So, as the name suggests, it is a complete Java development kit that includes everything including compiler, Java Runtime Environment (JRE), java debuggers, java docs, etc. For the program to execute in java, we need to install JDK on our computer in order to create, compile and run the java program.

4. Java Runtime Environment (JRE): JDK includes JRE. JRE installation on our computers allows the java program to run, however, we cannot compile it. JRE includes a browser, JVM, applet supports, and plugins. For running the java program, a computer needs JRE.

5. Garbage Collector: In Java, programmers can't delete the objects. To delete or recollect that memory JVM has a program called Garbage Collector. Garbage Collectors can recollect the objects that are not referenced. So Java makes the life of a programmer easy by handling memory management. However, programmers should be careful about their code whether they

are using objects that have been used for a long time. Because Garbage cannot recover the memory of objects being referenced.

6. ClassPath: The classpath is the file path where the java runtime and Java compiler look for **.class** files to load. By default, JDK provides many libraries. If you want to include external libraries they should be added to the classpath.

Primary/Main Features of Java

1. Platform Independent: Compiler converts source code to bytecode and then the JVM executes the bytecode generated by the compiler. This bytecode can run on any platform be it Windows, Linux, or macOS which means if we compile a program on Windows, then we can run it on Linux and vice versa. Each operating system has a different JVM, but the output produced by all the OS is the same after the execution of bytecode. That is why we call java a platform-independent language.

2. Object-Oriented Programming Language: Organizing the program in the terms of collection of objects is a way of object-oriented programming, each of which represents an instance of the class.

The four main concepts of Object-Oriented programming are:

- Abstraction
- Encapsulation
- Inheritance
- Polymorphism

3. Simple: Java is one of the simple languages as it does not have complex features like pointers, operator overloading, multiple inheritances, and Explicit memory allocation.

4. Robust: Java language is robust which means reliable. It is developed in such a way that it puts a lot of effort into checking errors as early as possible, that is why the java compiler is able to detect even those errors that are not easy to detect by another programming language. The main features of java that make it robust are garbage collection, Exception Handling, and memory allocation.

5. Secure: In java, we don't have pointers, so we cannot access out-of-bound arrays i.e it shows **ArrayIndexOutOfBound Exception** if we try to do so. That's why several security flaws like stack corruption or buffer overflow are impossible to exploit in Java. Also java programs run in an environment that is independent of the os(operating system) environment which makes java programs more secure .

6. Distributed: We can create distributed applications using the java programming language. Remote Method Invocation and Enterprise Java Beans are used for creating distributed applications in java. The java programs can be easily distributed on one or more systems that are connected to each other through an internet connection.

7. Multithreading: Java supports multithreading. It is a Java feature that allows concurrent execution of two or more parts of a program for maximum utilization of the CPU.

8. Portable: As we know, java code written on one machine can be run on another machine. The platform-independent feature of java in which its platform-independent bytecode can be taken to any platform for execution makes java portable.

9. High Performance: Java architecture is defined in such a way that it reduces overhead during the runtime and at some time java uses Just In Time (JIT) compiler where the compiler compiles code on-demand basics where it only compiles those methods that are called making applications to execute faster.

10. Dynamic flexibility: Java being completely object-oriented gives us the flexibility to add classes, new methods to existing classes and even create new classes through sub-classes. Java even supports functions written in other languages such as C, C++ which are referred to as native methods.

11. Sandbox Execution: Java programs run in a separate space that allows user to execute their applications without affecting the underlying system with help of a bytecode verifier. Bytecode verifier also provides additional security as its role is to check the code for any violation of access.

Data Gathering Documents

In research [1], the authors have developed an android application that can help the shoppers in searching for items in the store. the application has two screens corresponding to searching and map. The user can type in the products in the search bar. A centralized server returns available results for the search. On selecting the preferred search result, the map screen displays the location of the chosen item on the floor plan of the store.

In [2], we see how the authors have built an android application for an educational institute. The software android studio is used to develop the application. Android Studio has a large number of diverse views available for the development of the application. Some of the common views are button, scroll view, list view, checkbox, search view etc. A layout editor that supports drag and drop of components makes the app more user-friendly.

The research paper [3] describes a bookstore management application which is used to administrate the transaction of books. Users have to register and make an account during the first time login. A mobile application to evaluate English pronunciation skills is reported in research work [4]. In this app, the user is provided with random English words which are fetched from the SQLite Database. The user has to pronounce the words. The system compares the received input with the database and notifies whether the pronunciation is correct or incorrect.

In paper [5] the authors have developed an android application which can track the nutritional consumption of the users. The application was developed using Android Studio. The app can provide tips to the user for improving his/her diet.

In the research paper [6] the authors have introduced a location based application for shopping assistance. The users can search for nearby shops where a particular product is available. The location of the user is determined by the GPS (global positioning system) system of the mobile phone.

In [7] an Android application for controlling a robot is discussed. The application is used to achieve the movement of the robot. A Bluetooth module along with 8051 microcontroller is used to receive data from the smartphone.

In research [8] the authors introduce a wheelchair controlled by an android application. The application acts as a wireless control system for the wheelchair. The application is developed using the tool MIT app inventor. Bluetooth is used to connect the application to the wheelchair.

Paper [9] presents the implementation of an android app which functions as an user interface to control the movement of a pole climbing robot. In [10] the authors have given a detailed comparison on various platforms for android development. The platforms that have been reviewed are JAVA ME, .NET CF, Flash Lite, and Android. Various parameters like run time memory requirement, device support, User Interface support, development languages, deployment methods etc. are evaluated for these platforms.

In the paper, [11] authors identified the design requirements of a smart shopping cart which uses RFID readers and tags for the total price calculation. The RFID reader enables the user to pay the bill using the cart itself and thus avoids the tiring billing queues. The entire setup was made and they have also designed secure communication protocol to make the system more secure. A system capable of visualizing the data generated by biosensors for monitoring ICU patients is described in [12]. Roving's Bluetooth module RN-42 is used to transmit the data from the sensors to the android device. The android interface is programmed to reduce the noise in order to make the readings clearer in the paper [13].

For Mobile Shopping application SQLite is the perfect choice to create databases. It is an in-process library that implements a self-contained, serverless, zero-configuration; transactional SQL database engine which does not requires any database setup or administration.
Mobile Shopping application has one SQLite database in the SD card. It has all the items in the table. It has following tables:

1. *UserInfo table*: It contains the username and password.
2. *Product_Main table*: It contains all items available in store.
3. *Cart table*: It manages the list of items user want to purchase.
4. *Promotion table*: It contains all the discounts/offers in the store.
5. *Related product*: This table contains the records of related products.
6. *Storemaster table*: It contains the location details of all items in the store.

Here the SQLite database is created and all the operations are done on table using java interface.

E-R Diagram

Mobile Shopping application, Utterly is developed mainly using Android Studio.

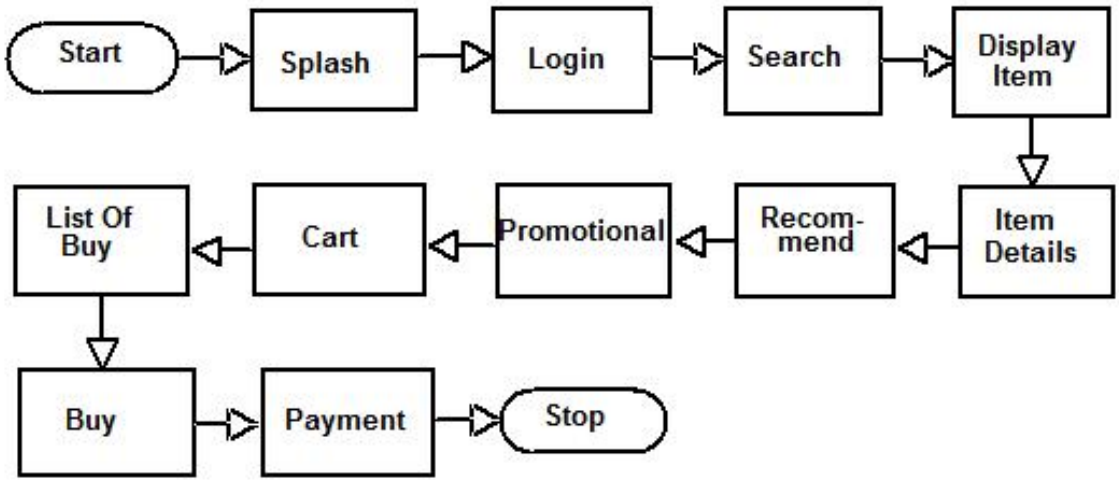

Fig. ER Diagram of Utterly

Source Code

Main Activity

```
package com.dhruva.shopping;

import android.app.ProgressDialog;
import android.content.Intent;

import android.os.Bundle;
import android.text.TextUtils;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

import com.dhruva.shopping.Model.Users;
import com.dhruva.shopping.Prevalent.Prevalent;
import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;

import io.paperdb.Paper;

public class MainActivity extends AppCompatActivity {
    private Button joinNowButton, loginButton;
    private ProgressDialog loadingBar;
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        joinNowButton = (Button) findViewById(R.id.main_join_now_btn);
        loginButton = (Button) findViewById(R.id.main_login_btn);
        loadingBar = new ProgressDialog(this);
        Paper.init(this);

        loginButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(MainActivity.this,
com.dhruva.shopping.LoginActivity.class);
                startActivity(intent);
            }
        });
        joinNowButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view)
            {
                Intent intent = new Intent(MainActivity.this, RegisterActivity.class);
                startActivity(intent);
            }
        });
        String UserPhoneKey = Paper.book().read(Prevalent.UserPhoneKey);
        String UserPasswordKey = Paper.book().read(Prevalent.UserPasswordKey);
        if (UserPhoneKey != "" && UserPasswordKey != "")
        {
            if (!TextUtils.isEmpty(UserPhoneKey)  &&  !TextUtils.isEmpty(UserPasswordKey))
            {
                AllowAccess(UserPhoneKey, UserPasswordKey);
```

```
            loadingBar.setTitle("Already Logged in");
            loadingBar.setMessage("Please wait.....");
            loadingBar.setCanceledOnTouchOutside(false);
            loadingBar.show();
        }
    }
}
private void AllowAccess(final String phone, final String password)
{
    final DatabaseReference RootRef;
    RootRef = FirebaseDatabase.getInstance().getReference();
    RootRef.addListenerForSingleValueEvent(new ValueEventListener() {
        @Override
        public void onDataChange(@NonNull DataSnapshot dataSnapshot) {
            if (dataSnapshot.child("Users").child(phone).exists()){

                Users usersData = dataSnapshot.child("Users").child(phone).getValue(Users.class);
                if (usersData.getPhone().equals(phone))
                {
                    if (usersData.getPassword().equals(password))
                    {
                        Toast.makeText(MainActivity.this, "Please wait, you are already logged in...",
Toast.LENGTH_SHORT).show();
                        loadingBar.dismiss();

                        Intent intent = new Intent(MainActivity.this, HomeActivity.class);
                        Prevalent.currentOnlineUser = usersData;
                        startActivity(intent);

                    }
```

```
                else {
                    loadingBar.dismiss();
                    Toast.makeText(MainActivity.this,"Password is
incorrect",Toast.LENGTH_SHORT).show();
                }
            }
        }
        else {
            Toast.makeText(MainActivity.this, "Account with this " + phone + " number do not
exists.", Toast.LENGTH_SHORT).show();
            loadingBar.dismiss();
        }
    }

    @Override
    public void onCancelled(DatabaseError databaseError) {

    }
  });
 }
}
```

Register Activity

```
package com.dhruva.shopping;

import android.app.ProgressDialog;
import android.content.Intent;

import android.os.Bundle;
```

```
import android.text.TextUtils;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

import com.google.android.gms.tasks.OnCompleteListener;
import com.google.android.gms.tasks.Task;
import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;

import java.util.HashMap;

public class RegisterActivity extends AppCompatActivity {
    private Button CreateAccountButton;
    private EditText InputName, InputPhoneNumber, InputPassword;
    private ProgressDialog loadingBar;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_register);
        CreateAccountButton = (Button) findViewById(R.id.register_btn);
        InputName = (EditText) findViewById(R.id.register_username_input);
        InputPassword = (EditText) findViewById(R.id.register_password_input);
        InputPhoneNumber = (EditText) findViewById(R.id.register_phone_number_input);
```

```
        loadingBar = new ProgressDialog(this);
        CreateAccountButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                CreateAccount();
            }
        });
    }
    private void CreateAccount(){
        String name = InputName.getText().toString();
        String phone = InputPhoneNumber.getText().toString();
        String password = InputPassword.getText().toString();
        if (TextUtils.isEmpty(name))
        {
            Toast.makeText(this, "Please write your name...", Toast.LENGTH_SHORT).show();
        }
        else if (TextUtils.isEmpty(phone))
        {
            Toast.makeText(this, "Please write your phone number...",
Toast.LENGTH_SHORT).show();
        }
        else if (TextUtils.isEmpty(password))
        {
            Toast.makeText(this, "Please write your password...", Toast.LENGTH_SHORT).show();
        }
        else
        {
            loadingBar.setTitle("Create Account");
            loadingBar.setMessage("Please wait, while we are checking the credentials.");
            loadingBar.setCanceledOnTouchOutside(false);
            loadingBar.show();
```

```
        ValidatephoneNumber(name, phone, password);
    }

  }

  private void ValidatephoneNumber(final String name, final String phone,final String password)
{
    final DatabaseReference RootRef;
    RootRef = FirebaseDatabase.getInstance().getReference();
    RootRef.addListenerForSingleValueEvent(new ValueEventListener() {
      @Override
      public void onDataChange(DataSnapshot dataSnapshot) {
        if (!(dataSnapshot.child("Users").child(phone).exists())){
          HashMap<String, Object> userdataMap = new HashMap<>();
          userdataMap.put("phone", phone);
          userdataMap.put("password", password);
          userdataMap.put("name", name);

RootRef.child("Users").child(phone).updateChildren(userdataMap).addOnCompleteListener(ne
w OnCompleteListener<Void>() {
            @Override
            public void onComplete(@NonNull Task<Void> task) {

              if (task.isSuccessful())
              {
                Toast.makeText(RegisterActivity.this, "Congratulations, your account has
been created.", Toast.LENGTH_SHORT).show();
                loadingBar.dismiss();
                Intent intent = new Intent(RegisterActivity.this,
com.dhruva.shopping.LoginActivity.class);
```

```
                    startActivity(intent);
                }
                else
                {
                    loadingBar.dismiss();
                    Toast.makeText(RegisterActivity.this, "Network Error: Please try again after
some time...", Toast.LENGTH_SHORT).show();
                }
            }
        });




        }
        else {
            Toast.makeText(RegisterActivity.this, "This " + phone + " already exists.",
Toast.LENGTH_SHORT).show();
            loadingBar.dismiss();
            Toast.makeText(RegisterActivity.this, "Please try again using another phone
number.", Toast.LENGTH_SHORT).show();
            Intent intent = new Intent(RegisterActivity.this,
com.dhruva.shopping.MainActivity.class);
            startActivity(intent);
        }
    }

    @Override
    public void onCancelled(DatabaseError databaseError) {

    }
});
```

```
    }
}
```

Login Activity

```
 public void onClick(View view)
        {
            LoginButton.setText("Login Admin");
            AdminLink.setVisibility(View.INVISIBLE);
            NotAdminLink.setVisibility(View.VISIBLE);
            parentDbName = "Admins";
        }
    });
    NotAdminLink.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view)
        {
            LoginButton.setText("Login");
            AdminLink.setVisibility(View.VISIBLE);
            NotAdminLink.setVisibility(View.INVISIBLE);
            parentDbName = "Users";
        }
    });

    }
    private void LoginUser()
    {
        String phone = InputPhoneNumber.getText().toString();
        String password = InputPassword.getText().toString();

        if (TextUtils.isEmpty(phone))
```

```
        {
            Toast.makeText(this, "Please write your phone number...",
Toast.LENGTH_SHORT).show();
        }
        else if (TextUtils.isEmpty(password))
        {
            Toast.makeText(this, "Please write your password...", Toast.LENGTH_SHORT).show();
        }
        else
        {
            loadingBar.setTitle("Login Account");
            loadingBar.setMessage("Please wait, while we are checking the credentials.");
            loadingBar.setCanceledOnTouchOutside(false);
            loadingBar.show();



            AllowAccessToAccount(phone, password);
        }
    }
    private void AllowAccessToAccount(final String phone, final String password)
    {

        if(chkBoxRememberMe.isChecked())
        {
            Paper.book().write(Prevalent.UserPhoneKey, phone);
            Paper.book().write(Prevalent.UserPasswordKey, password);
        }
        final DatabaseReference RootRef;
        RootRef = FirebaseDatabase.getInstance().getReference();
        RootRef.addListenerForSingleValueEvent(new ValueEventListener() {
            @Override
```

```
        public void onDataChange(@NonNull DataSnapshot dataSnapshot) {
            if (dataSnapshot.child(parentDbName).child(phone).exists()){

                Users usersData =
dataSnapshot.child(parentDbName).child(phone).getValue(Users.class);
                if (usersData.getPhone().equals(phone))
                {
                    if (usersData.getPassword().equals(password))
                    {
                        if(parentDbName.equals("Admins"))
                        {
                            Toast.makeText(LoginActivity.this, "Welcome Admin, you are logged in
Successfully...", Toast.LENGTH_SHORT).show();
                            loadingBar.dismiss();

                            Intent intent = new Intent(LoginActivity.this,
com.dhruva.shopping.AdminCategoryActivity.class);
                            startActivity(intent);
                        }
                        else if (parentDbName.equals("Users")){
                            Toast.makeText(LoginActivity.this, "logged in Successfully...",
Toast.LENGTH_SHORT).show();
                            loadingBar.dismiss();

                            Intent intent = new Intent(LoginActivity.this, HomeActivity.class);
                            Prevalent.currentOnlineUser = usersData;
                            startActivity(intent);
                        }

                    }
                    else {
```

```
                    loadingBar.dismiss();
                    Toast.makeText(LoginActivity.this,"Password is
incorrect",Toast.LENGTH_SHORT).show();
                }
            }
        }
        else {
            Toast.makeText(LoginActivity.this, "Account with this " + phone + " number do not
exists.", Toast.LENGTH_SHORT).show();
            loadingBar.dismiss();
        }
    }

    @Override
    public void onCancelled(DatabaseError databaseError) {

    }
  });
 }
}
```

Home Activity

```
package com.dhruva.shopping;

import androidx.annotation.NonNull;
import androidx.appcompat.app.ActionBarDrawerToggle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatDelegate;
import androidx.core.app.ActivityCompat;
import androidx.core.view.GravityCompat;
```

```
import androidx.drawerlayout.widget.DrawerLayout;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import android.Manifest;
import android.content.Intent;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.speech.RecognizerIntent;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.Menu;
import android.view.MenuItem;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ImageButton;
import android.widget.RelativeLayout;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.widget.Toolbar;

import com.dhruva.shopping.Model.Products;
import com.dhruva.shopping.Prevalent.Prevalent;
import com.dhruva.shopping.ViewHolder.ProductViewHolder;
import com.firebase.ui.database.FirebaseRecyclerAdapter;
import com.firebase.ui.database.FirebaseRecyclerOptions;
import com.google.android.material.floatingactionbutton.FloatingActionButton;
import com.google.android.material.navigation.NavigationView;
```

```
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.squareup.picasso.Picasso;


import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;
import java.util.Objects;

import de.hdodenhof.circleimageview.CircleImageView;
import io.paperdb.Paper;

public class HomeActivity extends AppCompatActivity implements
NavigationView.OnNavigationItemSelectedListener {
    private static final int REQUEST_CODE_SPEECH_INPUT = 1;
    private DatabaseReference ProductsRef;
    DrawerLayout drawerLayout;
    NavigationView navigationView;
    Toolbar toolbar;
    TextView textView;
    private RecyclerView recyclerView;
    RecyclerView.LayoutManager layoutManager;
    Button grocery, electronics, pharmacy, foodItems;


    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_home);
```

```
grocery = findViewById(R.id.grocery);
electronics = findViewById(R.id.electronics);
pharmacy = findViewById(R.id.pharmacy);
foodItems = findViewById(R.id.food);
ImageButton speakButton = (ImageButton) findViewById(R.id.speakButton);



drawerLayout = findViewById(R.id.drawer_layout);
navigationView = findViewById(R.id.nav_view);
toolbar = findViewById(R.id.toolbar);
//androidx.appcompat.widget.Toolbar toolbar = (androidx.appcompat.widget.Toolbar)
findViewById(R.id.toolbar);
toolbar.setTitle("Home");
//  setSupportActionBar(toolbar);

navigationView.bringToFront();
ActionBarDrawerToggle toggle = new ActionBarDrawerToggle(this, drawerLayout, toolbar,
R.string.navigation_drawer_open, R.string.navigation_drawer_close);
drawerLayout.addDrawerListener(toggle);
toggle.syncState();
navigationView.setNavigationItemSelectedListener(this);



/* View headerView = navigationView.getHeaderView(0);
TextView userNameTextView = headerView.findViewById(R.id.user_profile_name);
CircleImageView profileImageView = headerView.findViewById(R.id.user_profile_image);
userNameTextView.setText(Prevalent.currentOnlineUser.getName());

Picasso.get().load(Prevalent.currentOnlineUser.getImage()).placeholder(R.drawable.profile).into
(profileImageView);
recyclerView = findViewById(R.id.recycler_menu);
```

```
recyclerView.setHasFixedSize(true);
layoutManager = new LinearLayoutManager(this);
recyclerView.setLayoutManager(layoutManager);*/

grocery.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Intent intent = new Intent(HomeActivity.this, grocery.class);
        startActivity(intent);
    }
});

electronics.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Intent intent = new Intent(HomeActivity.this, electronics.class);
        startActivity(intent);
    }
});

pharmacy.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Intent intent = new Intent(HomeActivity.this, pharmacy.class);
        startActivity(intent);
    }
});

foodItems.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
```

```
            Intent intent = new Intent(HomeActivity.this, foodItems.class);
            startActivity(intent);
        }
    });

    FloatingActionButton fab = (FloatingActionButton) findViewById(R.id.fab);
    fab.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            Intent intent = new Intent(HomeActivity.this, CartActivity.class);
            startActivity(intent);
        }
    });


}

public void speakButtonClicked(View v) {
    startVoiceRecognitionActivity();
}

@Override
public void onBackPressed() {
    if (drawerLayout.isDrawerOpen(GravityCompat.START)) {
        drawerLayout.closeDrawer(GravityCompat.START);
    } else {
        super.onBackPressed();
    }
}

@Override
```

```
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar if it is present.
        getMenuInflater().inflate(R.menu.main_menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        // Handle action bar item clicks here. The action bar will
        // automatically handle clicks on the Home/Up button, so long
        // as you specify a parent activity in AndroidManifest.xml.
        int id = item.getItemId();

        //noinspection SimplifiableIfStatement
//        if (id == R.id.action_settings) {
//            return true;
//        }

        return super.onOptionsItemSelected(item);
    }


    private void startVoiceRecognitionActivity() {
        Intent intent = new Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH);
        intent.putExtra(RecognizerIntent.EXTRA_LANGUAGE_MODEL,
                RecognizerIntent.LANGUAGE_MODEL_FREE_FORM);
        intent.putExtra(RecognizerIntent.EXTRA_PROMPT, "Say your category...");
        intent.putExtra(RecognizerIntent.EXTRA_MAX_RESULTS, 1);

        startActivityForResult(intent, REQUEST_CODE_SPEECH_INPUT);
    }
```

```
    @Override
    public boolean onNavigationItemSelected(MenuItem item) {
        // Handle navigation view item clicks here.
        int id = item.getItemId();

        if (id == R.id.nav_cart) {
            Intent intent = new Intent(HomeActivity.this, CartActivity.class);
            startActivity(intent);
        } else if (id == R.id.nav_search) {
            Intent intent = new Intent(HomeActivity.this, SearchProductsActivity.class);
            startActivity(intent);

        } else if (id == R.id.nav_categories) {

        } else if (id == R.id.nav_settings) {
            Intent intent = new Intent(HomeActivity.this, SettinsActivity.class);
            startActivity(intent);

        } else if (id == R.id.nav_logout) {
            Paper.book().destroy();
            Intent intent = new Intent(HomeActivity.this, MainActivity.class);
            intent.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TASK);
            startActivity(intent);
            finish();

        }


        drawerLayout.closeDrawer(GravityCompat.START);
```

```
        return true;
    }


    @Override
    protected void onActivityResult(int requestCode, int resultCode, Intent data) {
        String tv_Speech_to_text = "";
        super.onActivityResult(requestCode, resultCode, data);

        if(requestCode == REQUEST_CODE_SPEECH_INPUT)
        {
            if(resultCode == RESULT_OK && data != null)  {
                ArrayList<String> result = data.getStringArrayListExtra(
                        RecognizerIntent.EXTRA_RESULTS);
                tv_Speech_to_text = Objects.requireNonNull((result).get(0));
                Toast.makeText(this, tv_Speech_to_text, Toast.LENGTH_SHORT).show();
                if(tv_Speech_to_text.equals("grocery")) {
                    Intent intent = new Intent(HomeActivity.this, grocery.class);
                    startActivity(intent);
                }
                else if(tv_Speech_to_text.equals("electronics")) {
                    Intent intent = new Intent(HomeActivity.this, electronics.class);
                    startActivity(intent);
                }
                else if(tv_Speech_to_text.equals("pharmacy") || tv_Speech_to_text.equals("medicine"))
{
                    Intent intent = new Intent(HomeActivity.this, pharmacy.class);
                    startActivity(intent);
                }
                else if(tv_Speech_to_text.equals("food") || tv_Speech_to_text.equals("food items")) {
                    Intent intent = new Intent(HomeActivity.this, foodItems.class);
```

```
                startActivity(intent);
            }
            else if(tv_Speech_to_text.equals("cart")|| tv_Speech_to_text.equals("karte")) {
                Intent intent = new Intent(HomeActivity.this,CartActivity.class);
                startActivity(intent);
            }
            }
        }


    }


}
```

Grocery

```
package com.dhruva.shopping;

import android.content.Intent;
import android.os.Bundle;
import android.speech.RecognizerIntent;
import android.view.LayoutInflater;
import android.view.Menu;
import android.view.MenuItem;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageButton;
import android.widget.TextView;
import android.widget.Toast;
```

```
import androidx.annotation.NonNull;
import androidx.appcompat.app.ActionBarDrawerToggle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatDelegate;
import androidx.appcompat.widget.Toolbar;
import androidx.core.view.GravityCompat;
import androidx.drawerlayout.widget.DrawerLayout;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.dhruva.shopping.Model.Products;
import com.dhruva.shopping.Prevalent.Prevalent;
import com.dhruva.shopping.ViewHolder.ProductViewHolder;
import com.firebase.ui.database.FirebaseRecyclerAdapter;
import com.firebase.ui.database.FirebaseRecyclerOptions;
import com.google.android.material.floatingactionbutton.FloatingActionButton;
import com.google.android.material.navigation.NavigationView;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.squareup.picasso.Picasso;

import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;
import java.util.Objects;

import de.hdodenhof.circleimageview.CircleImageView;
import io.paperdb.Paper;

public class grocery extends AppCompatActivity implements
NavigationView.OnNavigationItemSelectedListener {
```

```
    private static final int REQUEST_CODE_SPEECH_INPUT = 1;
    private DatabaseReference ProductsRef;
    DrawerLayout drawerLayout;
    NavigationView navigationView;
    Toolbar toolbar;
    TextView textView;
    private RecyclerView recyclerView;
    RecyclerView.LayoutManager layoutManager;

    List<String> gList =  Arrays.asList("Johnson Baby Soap","Surf Excel","Kissan Jam","Toor
Dal","Tata Salt");

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_grocery);
        AppCompatDelegate.setDefaultNightMode(AppCompatDelegate.MODE_NIGHT_YES);
        ProductsRef = FirebaseDatabase.getInstance().getReference().child("Products");

        ImageButton speakButton = (ImageButton) findViewById(R.id.speakButton);
        drawerLayout = findViewById(R.id.drawer_layout1);
        navigationView = findViewById(R.id.nav_view1);
        toolbar = findViewById(R.id.category_toolbar);
//        Toolbar toolbar = (Toolbar) findViewById(R.id.category_toolbar);
        toolbar.setTitle("Home");
//        setSupportActionBar(toolbar);

        navigationView.bringToFront();
        ActionBarDrawerToggle toggle = new ActionBarDrawerToggle(this, drawerLayout, toolbar,
R.string.navigation_drawer_open, R.string.navigation_drawer_close);
        drawerLayout.addDrawerListener(toggle);
```

```
        toggle.syncState();
        navigationView.setNavigationItemSelectedListener(this);



        View headerView = navigationView.getHeaderView(0);
        TextView userNameTextView = headerView.findViewById(R.id.user_profile_name);
        CircleImageView profileImageView = headerView.findViewById(R.id.user_profile_image);
        userNameTextView.setText(Prevalent.currentOnlineUser.getName());

Picasso.get().load(Prevalent.currentOnlineUser.getImage()).placeholder(R.drawable.profile).into
(profileImageView);
        recyclerView = findViewById(R.id.recycler_menu);
        recyclerView.setHasFixedSize(true);
        layoutManager = new LinearLayoutManager(this);
        recyclerView.setLayoutManager(layoutManager);

        FloatingActionButton fab = (FloatingActionButton) findViewById(R.id.fab1);
        fab.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(grocery.this, CartActivity.class);
                startActivity(intent);
            }
        });



    }

    public void speakButtonClicked(View v) {
        startVoiceRecognitionActivity();
    }
```

```
    @Override
    protected void onStart() {
        super.onStart();

        FirebaseRecyclerOptions<Products> options =
                new FirebaseRecyclerOptions.Builder<Products>()
                        .setQuery(ProductsRef, Products.class)
                        .build();

        FirebaseRecyclerAdapter<Products, ProductViewHolder> adapter =
                new FirebaseRecyclerAdapter<Products, ProductViewHolder>(options) {
                    /*for (int i=0;i < myString.size();i++)
                    {
                        gList.add(model.getPid());
                    }*/
                    @Override
                    protected void onBindViewHolder(@NonNull ProductViewHolder holder, int
position, @NonNull final Products model) {
                        holder.txtProductName.setText(model.getPname());
                        holder.txtProductDescription.setText(model.getDescription());
                        holder.txtProductPrice.setText("Price = " + model.getPrice() + "Rs.");
                        Picasso.get().load(model.getImage()).into(holder.imageView);
                        holder.itemView.setOnClickListener(new View.OnClickListener() {
                            @Override
                            public void onClick(View view) {
                                Intent intent = new Intent(grocery.this, ProductDetailsActivity.class);
                                intent.putExtra("pid", model.getPid());
                                startActivity(intent);
                            }
                        });
```

```
            }

            @NonNull
            @Override
            public ProductViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int
viewType) {
                View view =
LayoutInflater.from(parent.getContext()).inflate(R.layout.product_items_layout, parent, false);
                ProductViewHolder holder = new ProductViewHolder(view);
                return holder;
            }
        };
    recyclerView.setAdapter(adapter);
    adapter.startListening();

  }

  @Override
  public void onBackPressed() {
    if (drawerLayout.isDrawerOpen(GravityCompat.START)) {
      drawerLayout.closeDrawer(GravityCompat.START);
    } else {
      super.onBackPressed();
    }
  }

  @Override
  public boolean onCreateOptionsMenu(Menu menu) {
    // Inflate the menu; this adds items to the action bar if it is present.
    getMenuInflater().inflate(R.menu.main_menu, menu);
    return true;
```

```
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        // Handle action bar item clicks here. The action bar will
        // automatically handle clicks on the Home/Up button, so long
        // as you specify a parent activity in AndroidManifest.xml.
        int id = item.getItemId();

        //noinspection SimplifiableIfStatement
//      if (id == R.id.action_settings) {
//          return true;
//      }

        return super.onOptionsItemSelected(item);
    }

    private void startVoiceRecognitionActivity() {
        Intent intent = new Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH);
        intent.putExtra(RecognizerIntent.EXTRA_LANGUAGE_MODEL,
                RecognizerIntent.LANGUAGE_MODEL_FREE_FORM);
        intent.putExtra(RecognizerIntent.EXTRA_PROMPT, "Say your Product...");
        intent.putExtra(RecognizerIntent.EXTRA_MAX_RESULTS, 1);

        startActivityForResult(intent, REQUEST_CODE_SPEECH_INPUT);
    }


    @Override
    public boolean onNavigationItemSelected(MenuItem item) {
        // Handle navigation view item clicks here.
```

```
        int id = item.getItemId();

        if (id == R.id.nav_cart) {
            Intent intent = new Intent(grocery.this, CartActivity.class);
            startActivity(intent);
        } else if (id == R.id.nav_search) {
            Intent intent = new Intent(grocery.this, SearchProductsActivity.class);
            startActivity(intent);

        } else if (id == R.id.nav_categories) {

        } else if (id == R.id.nav_settings) {
            Intent intent = new Intent(grocery.this, SettinsActivity.class);
            startActivity(intent);

        } else if (id == R.id.nav_logout) {
            Paper.book().destroy();
            Intent intent = new Intent(grocery.this, MainActivity.class);
            intent.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TASK);
            startActivity(intent);
            finish();

        }



        drawerLayout.closeDrawer(GravityCompat.START);
        return true;
    }
```

```
@Override
protected void onActivityResult(int requestCode, int resultCode, Intent data) {
    String tv_Speech_to_text = "";
    String parameter = "";
    super.onActivityResult(requestCode, resultCode, data);

    if (requestCode == REQUEST_CODE_SPEECH_INPUT) {
        if (resultCode == RESULT_OK && data != null) {
            ArrayList<String> result = data.getStringArrayListExtra(
                    RecognizerIntent.EXTRA_RESULTS);
            tv_Speech_to_text = Objects.requireNonNull((result).get(0));

            if (tv_Speech_to_text.equals("cart")|| tv_Speech_to_text.equals("karte")) {
                Intent intent = new Intent(grocery.this,CartActivity.class);
                startActivity(intent);
            }
            } else if(tv_Speech_to_text.equals("category")|| tv_Speech_to_text.equals("back")) {
            Toast.makeText(this, "back pressed", Toast.LENGTH_SHORT).show();
            super.onBackPressed();
            }
                for(String item:gList)
                {
                    if(tv_Speech_to_text.equals(item)) {
                        Toast.makeText(this, item, Toast.LENGTH_SHORT).show();

                        if(item.equals("Johnson Baby Soap"))
                            parameter = "Jul 30, 202223:15:17 pm";
                        else if(item.equals("Surf Excel"))
                            parameter = "Jul 30, 202223:21:34 pm";
                        else if(item.equals("Kissan Jam"))
                            parameter = "Jul 30, 202223:23:49 pm";
```

```
                else if(item.equals("Toor Dal"))
                    parameter = "Jul 30, 202223:25:06 pm";
                else if(item.equals("Tata Salt"))
                    parameter = "Jul 30, 202223:26:48 pm";
                Intent intent = new Intent(grocery.this, ProductDetailsActivity.class);
                intent.putExtra("pid", parameter);
                startActivity(intent);
            }
        }




    }
  }
}
```

Electronics

```
package com.dhruva.shopping;

import android.content.Intent;
import android.os.Bundle;
import android.speech.RecognizerIntent;
import android.view.LayoutInflater;
import android.view.Menu;
import android.view.MenuItem;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
```

```
import androidx.appcompat.app.ActionBarDrawerToggle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatDelegate;
import androidx.appcompat.widget.Toolbar;
import androidx.core.view.GravityCompat;
import androidx.drawerlayout.widget.DrawerLayout;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.dhruva.shopping.Model.Products;
import com.dhruva.shopping.Prevalent.Prevalent;
import com.dhruva.shopping.ViewHolder.ProductViewHolder;
import com.firebase.ui.database.FirebaseRecyclerAdapter;
import com.firebase.ui.database.FirebaseRecyclerOptions;
import com.google.android.material.floatingactionbutton.FloatingActionButton;
import com.google.android.material.navigation.NavigationView;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.squareup.picasso.Picasso;

import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;
import java.util.Objects;

import de.hdodenhof.circleimageview.CircleImageView;
import io.paperdb.Paper;

public class electronics extends AppCompatActivity implements
NavigationView.OnNavigationItemSelectedListener {
    private static final int REQUEST_CODE_SPEECH_INPUT = 1;
```

```
    private DatabaseReference ProductsRef;
    DrawerLayout drawerLayout;
    NavigationView navigationView;
    Toolbar toolbar;
    TextView textView;
    private RecyclerView recyclerView;
    RecyclerView.LayoutManager layoutManager;

    List<String> eList =  Arrays.asList("Sony headphones","wireless mouse","Apple
iPhone","washing machine","laptop");

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_grocery);
        AppCompatDelegate.setDefaultNightMode(AppCompatDelegate.MODE_NIGHT_YES);
        ProductsRef = FirebaseDatabase.getInstance().getReference().child("Electronics");

       drawerLayout=findViewById(R.id.drawer_layout1);
        navigationView=findViewById(R.id.nav_view1);
        toolbar=findViewById(R.id.category_toolbar);
//        Toolbar toolbar = (Toolbar) findViewById(R.id.category_toolbar);
       toolbar.setTitle("Home");
//        setSupportActionBar(toolbar);

        navigationView.bringToFront();
        ActionBarDrawerToggle toggle=new
ActionBarDrawerToggle(this,drawerLayout,toolbar,R.string.navigation_drawer_open,R.string.na
vigation_drawer_close);
        drawerLayout.addDrawerListener(toggle);
        toggle.syncState();
```

```
        navigationView.setNavigationItemSelectedListener(this);



        View headerView = navigationView.getHeaderView(0);
        TextView userNameTextView = headerView.findViewById(R.id.user_profile_name);
        CircleImageView profileImageView = headerView.findViewById(R.id.user_profile_image);
        userNameTextView.setText(Prevalent.currentOnlineUser.getName());

Picasso.get().load(Prevalent.currentOnlineUser.getImage()).placeholder(R.drawable.profile).into
(profileImageView);
        recyclerView = findViewById(R.id.recycler_menu);
        recyclerView.setHasFixedSize(true);
        layoutManager = new LinearLayoutManager(this);
        recyclerView.setLayoutManager(layoutManager);

        FloatingActionButton fab = (FloatingActionButton) findViewById(R.id.fab1);
        fab.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(electronics.this,CartActivity.class);
                startActivity(intent);
            }
        });

    }

    public void speakButtonClicked(View v) {
        startVoiceRecognitionActivity();
    }

    @Override
```

```
    protected void onStart() {
        super.onStart();
        FirebaseRecyclerOptions<Products> options =
                new FirebaseRecyclerOptions.Builder<Products>()
                        .setQuery(ProductsRef, Products.class)
                        .build();

        FirebaseRecyclerAdapter<Products, ProductViewHolder> adapter =
                new FirebaseRecyclerAdapter<Products, ProductViewHolder>(options) {
                    @Override
                    protected void onBindViewHolder(@NonNull ProductViewHolder holder, int
position, @NonNull final Products model)
                    {
                        holder.txtProductName.setText(model.getPname());
                        holder.txtProductDescription.setText(model.getDescription());
                        holder.txtProductPrice.setText("Price = " + model.getPrice() + "Rs.");
                        Picasso.get().load(model.getImage()).into(holder.imageView);
                        holder.itemView.setOnClickListener(new View.OnClickListener() {
                            @Override
                            public void onClick(View view) {
                                Intent intent =new Intent(electronics.this,ElecProductDetailsActivity.class);
                                intent.putExtra("pid",model.getPid());
                                startActivity(intent);
                            }
                        });
                    }

                    @NonNull
                    @Override
                    public ProductViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int
viewType) {
```

```
                View view =
LayoutInflater.from(parent.getContext()).inflate(R.layout.product_items_layout, parent, false);
                ProductViewHolder holder = new ProductViewHolder(view);
                return holder;
            }
        };
    recyclerView.setAdapter(adapter);
    adapter.startListening();

}
@Override
public void onBackPressed(){
    if(drawerLayout.isDrawerOpen(GravityCompat.START)){
        drawerLayout.closeDrawer(GravityCompat.START);
    }
    else
    {super.onBackPressed();
    }
}

@Override
public boolean onCreateOptionsMenu(Menu menu) {
    // Inflate the menu; this adds items to the action bar if it is present.
    getMenuInflater().inflate(R.menu.main_menu, menu);
    return true;
}

@Override
public boolean onOptionsItemSelected(MenuItem item) {
    // Handle action bar item clicks here. The action bar will
    // automatically handle clicks on the Home/Up button, so long
```

```
        // as you specify a parent activity in AndroidManifest.xml.
        int id = item.getItemId();

        //noinspection SimplifiableIfStatement
//        if (id == R.id.action_settings) {
//            return true;
//        }

        return super.onOptionsItemSelected(item);
    }




    @Override
    public boolean onNavigationItemSelected(MenuItem item) {
        // Handle navigation view item clicks here.
        int id = item.getItemId();

        if (id == R.id.nav_cart) {
            Intent intent = new Intent(electronics.this,CartActivity.class);
            startActivity(intent);
        } else if (id == R.id.nav_search) {
            Intent intent = new Intent(electronics.this,SearchProductsActivity.class);
            startActivity(intent);

        } else if (id == R.id.nav_categories) {

        } else if (id == R.id.nav_settings) {
            Intent intent=new Intent(electronics.this,SettinsActivity.class);
            startActivity(intent);
```

```
        } else if (id == R.id.nav_logout) {
            Paper.book().destroy();
            Intent intent=new Intent(electronics.this,MainActivity.class);
            intent.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK
|Intent.FLAG_ACTIVITY_CLEAR_TASK );
            startActivity(intent);
            finish();

        }


        drawerLayout.closeDrawer(GravityCompat.START);
        return true;
    }

    private void startVoiceRecognitionActivity() {
        Intent intent = new Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH);
        intent.putExtra(RecognizerIntent.EXTRA_LANGUAGE_MODEL,
                RecognizerIntent.LANGUAGE_MODEL_FREE_FORM);
        intent.putExtra(RecognizerIntent.EXTRA_PROMPT, "Say your Product...");
        intent.putExtra(RecognizerIntent.EXTRA_MAX_RESULTS, 1);

        startActivityForResult(intent, REQUEST_CODE_SPEECH_INPUT);
    }


    @Override
    protected void onActivityResult(int requestCode, int resultCode, Intent data) {
        String tv_Speech_to_text = "";
        String parameter = "";
```

```
super.onActivityResult(requestCode, resultCode, data);

if (requestCode == REQUEST_CODE_SPEECH_INPUT) {
   if (resultCode == RESULT_OK && data != null) {
      ArrayList<String> result = data.getStringArrayListExtra(
            RecognizerIntent.EXTRA_RESULTS);
      tv_Speech_to_text = Objects.requireNonNull((result).get(0));

      if (tv_Speech_to_text.equals("cart")|| tv_Speech_to_text.equals("karte")) {
         Intent intent = new Intent(electronics.this,CartActivity.class);
         startActivity(intent);
      }
   } else if(tv_Speech_to_text.equals("category")|| tv_Speech_to_text.equals("back")) {
      Toast.makeText(this, "back pressed", Toast.LENGTH_SHORT).show();
      super.onBackPressed();
   }
   for(String item:eList)
   {
      if(tv_Speech_to_text.equals(item)) {
         Toast.makeText(this, item, Toast.LENGTH_SHORT).show();

         if(item.equals("Sony headphones"))
            parameter = "Jul 31, 202200:13:18 am";
         else if(item.equals("wireless mouse"))
            parameter = "Jul 31, 202200:13:55 am";
         else if(item.equals("Apple iPhone"))
            parameter = "Jul 31, 202200:14:33 am";
         else if(item.equals("washing machine"))
            parameter = "Jul 31, 202200:15:34 am";
         else if(item.equals("laptop"))
            parameter = "Jul 31, 202212:26:22 pm";
```

```
                Intent intent = new Intent(electronics.this, ElecProductDetailsActivity.class);
                intent.putExtra("pid", parameter);
                startActivity(intent);
            }
        }

    }
}
```

Pharmacy

```
package com.dhruva.shopping;

import android.content.Intent;
import android.os.Bundle;
import android.speech.RecognizerIntent;
import android.view.LayoutInflater;
import android.view.Menu;
import android.view.MenuItem;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.ActionBarDrawerToggle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatDelegate;
import androidx.appcompat.widget.Toolbar;
import androidx.core.view.GravityCompat;
import androidx.drawerlayout.widget.DrawerLayout;
```

```
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.dhruva.shopping.Model.Products;
import com.dhruva.shopping.Prevalent.Prevalent;
import com.dhruva.shopping.ViewHolder.ProductViewHolder;
import com.firebase.ui.database.FirebaseRecyclerAdapter;
import com.firebase.ui.database.FirebaseRecyclerOptions;
import com.google.android.material.floatingactionbutton.FloatingActionButton;
import com.google.android.material.navigation.NavigationView;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.squareup.picasso.Picasso;

import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;
import java.util.Objects;

import de.hdodenhof.circleimageview.CircleImageView;
import io.paperdb.Paper;

public class pharmacy extends AppCompatActivity implements
NavigationView.OnNavigationItemSelectedListener {
    private static final int REQUEST_CODE_SPEECH_INPUT = 1;
    private DatabaseReference ProductsRef;
    DrawerLayout drawerLayout;
    NavigationView navigationView;
    Toolbar toolbar;
    TextView textView;
    private RecyclerView recyclerView;
```

```
    RecyclerView.LayoutManager layoutManager;

    List<String> mList =  Arrays.asList("metformin","baby rub","vaporizer","baby gift
pack","Dolo 650");
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_grocery);
        AppCompatDelegate.setDefaultNightMode(AppCompatDelegate.MODE_NIGHT_YES);
        ProductsRef = FirebaseDatabase.getInstance().getReference().child("Pharmacy");

       drawerLayout=findViewById(R.id.drawer_layout1);
        navigationView=findViewById(R.id.nav_view1);
        toolbar=findViewById(R.id.category_toolbar);
//        Toolbar toolbar = (Toolbar) findViewById(R.id.category_toolbar);
       toolbar.setTitle("Home");
//        setSupportActionBar(toolbar);

        navigationView.bringToFront();
        ActionBarDrawerToggle toggle=new
ActionBarDrawerToggle(this,drawerLayout,toolbar,R.string.navigation_drawer_open,R.string.na
vigation_drawer_close);
        drawerLayout.addDrawerListener(toggle);
        toggle.syncState();
        navigationView.setNavigationItemSelectedListener(this);


        View headerView = navigationView.getHeaderView(0);
        TextView userNameTextView = headerView.findViewById(R.id.user_profile_name);
        CircleImageView profileImageView = headerView.findViewById(R.id.user_profile_image);
        userNameTextView.setText(Prevalent.currentOnlineUser.getName());
```

```
Picasso.get().load(Prevalent.currentOnlineUser.getImage()).placeholder(R.drawable.profile).into
(profileImageView);
        recyclerView = findViewById(R.id.recycler_menu);
        recyclerView.setHasFixedSize(true);
        layoutManager = new LinearLayoutManager(this);
        recyclerView.setLayoutManager(layoutManager);

        FloatingActionButton fab = (FloatingActionButton) findViewById(R.id.fab1);
        fab.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(pharmacy.this,CartActivity.class);
                startActivity(intent);
            }
        });

    }

    public void speakButtonClicked(View v) {
        startVoiceRecognitionActivity();
    }
   @Override
    protected void onStart() {
        super.onStart();
        FirebaseRecyclerOptions<Products> options =
                new FirebaseRecyclerOptions.Builder<Products>()
                        .setQuery(ProductsRef, Products.class)
                        .build();

        FirebaseRecyclerAdapter<Products, ProductViewHolder> adapter =
```

```
            new FirebaseRecyclerAdapter<Products, ProductViewHolder>(options) {
                @Override
                protected void onBindViewHolder(@NonNull ProductViewHolder holder, int
position, @NonNull final Products model)
                {
                    holder.txtProductName.setText(model.getPname());
                    holder.txtProductDescription.setText(model.getDescription());
                    holder.txtProductPrice.setText("Price = " + model.getPrice() + "Rs.");
                    Picasso.get().load(model.getImage()).into(holder.imageView);
                    holder.itemView.setOnClickListener(new View.OnClickListener() {
                        @Override
                        public void onClick(View view) {
                            Intent intent =new Intent(pharmacy.this,MedProductDetailsActivity.class);
                            intent.putExtra("pid",model.getPid());
                            startActivity(intent);
                        }
                    });
                }

                @NonNull
                @Override
                public ProductViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int
viewType) {
                    View view =
LayoutInflater.from(parent.getContext()).inflate(R.layout.product_items_layout, parent, false);
                    ProductViewHolder holder = new ProductViewHolder(view);
                    return holder;
                }
            };
        recyclerView.setAdapter(adapter);
        adapter.startListening();
```

```
    }
    @Override
    public void onBackPressed(){
        if(drawerLayout.isDrawerOpen(GravityCompat.START)){
            drawerLayout.closeDrawer(GravityCompat.START);
        }
        else
        {super.onBackPressed();
        }
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar if it is present.
        getMenuInflater().inflate(R.menu.main_menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        // Handle action bar item clicks here. The action bar will
        // automatically handle clicks on the Home/Up button, so long
        // as you specify a parent activity in AndroidManifest.xml.
        int id = item.getItemId();

        //noinspection SimplifiableIfStatement
//        if (id == R.id.action_settings) {
//            return true;
//        }
```

```
        return super.onOptionsItemSelected(item);
    }





    @Override
    public boolean onNavigationItemSelected(MenuItem item) {
        // Handle navigation view item clicks here.
        int id = item.getItemId();

        if (id == R.id.nav_cart) {
            Intent intent = new Intent(pharmacy.this,CartActivity.class);
            startActivity(intent);
        } else if (id == R.id.nav_search) {
            Intent intent = new Intent(pharmacy.this,SearchProductsActivity.class);
            startActivity(intent);

        } else if (id == R.id.nav_categories) {

        } else if (id == R.id.nav_settings) {
            Intent intent=new Intent(pharmacy.this,SettinsActivity.class);
            startActivity(intent);

        } else if (id == R.id.nav_logout) {
            Paper.book().destroy();
            Intent intent=new Intent(pharmacy.this,MainActivity.class);
            intent.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK
|Intent.FLAG_ACTIVITY_CLEAR_TASK );
            startActivity(intent);
            finish();
```

```
    }


    drawerLayout.closeDrawer(GravityCompat.START);
    return true;
}

private void startVoiceRecognitionActivity() {
    Intent intent = new Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH);
    intent.putExtra(RecognizerIntent.EXTRA_LANGUAGE_MODEL,
            RecognizerIntent.LANGUAGE_MODEL_FREE_FORM);
    intent.putExtra(RecognizerIntent.EXTRA_PROMPT, "Say your Product...");
    intent.putExtra(RecognizerIntent.EXTRA_MAX_RESULTS, 1);

    startActivityForResult(intent, REQUEST_CODE_SPEECH_INPUT);
}


@Override
protected void onActivityResult(int requestCode, int resultCode, Intent data) {
    String tv_Speech_to_text = "";
    String parameter = "";
    super.onActivityResult(requestCode, resultCode, data);

    if (requestCode == REQUEST_CODE_SPEECH_INPUT) {
        if (resultCode == RESULT_OK && data != null) {
            ArrayList<String> result = data.getStringArrayListExtra(
                    RecognizerIntent.EXTRA_RESULTS);
            tv_Speech_to_text = Objects.requireNonNull((result).get(0));
```

```
            if (tv_Speech_to_text.equals("cart") || tv_Speech_to_text.equals("karte")) {
                Intent intent = new Intent(pharmacy.this, CartActivity.class);
                startActivity(intent);
            }
        } else if (tv_Speech_to_text.equals("category") || tv_Speech_to_text.equals("back")) {
            Toast.makeText(this, "back pressed", Toast.LENGTH_SHORT).show();
            super.onBackPressed();
        }

    }
    for (String item : mList) {
        if (tv_Speech_to_text.equals(item)) {
            Toast.makeText(this, item, Toast.LENGTH_SHORT).show();

            if (item.equals("metformin"))
                parameter = "Jul 31, 202200:02:39 am";
            else if (item.equals("baby rub"))
                parameter = "Jul 31, 202200:09:07 am";
            else if (item.equals("vaporizer"))
                parameter = "Jul 31, 202200:10:46 am";
            else if (item.equals("baby gift pack"))
                parameter = "Jul 31, 202200:21:16 am";
            else if (item.equals("Dolo 650"))
                parameter = "Jul 31, 202200:22:47 am";
            Intent intent = new Intent(pharmacy.this, MedProductDetailsActivity.class);
            intent.putExtra("pid", parameter);
            startActivity(intent);
        }
    }

}
```

```
}
```

Food Items

```
package com.dhruva.shopping;

import android.content.Intent;
import android.os.Bundle;
import android.speech.RecognizerIntent;
import android.view.LayoutInflater;
import android.view.Menu;
import android.view.MenuItem;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.ActionBarDrawerToggle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.app.AppCompatDelegate;
import androidx.appcompat.widget.Toolbar;
import androidx.core.view.GravityCompat;
import androidx.drawerlayout.widget.DrawerLayout;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import com.dhruva.shopping.Model.Products;
import com.dhruva.shopping.Prevalent.Prevalent;
import com.dhruva.shopping.ViewHolder.ProductViewHolder;
import com.firebase.ui.database.FirebaseRecyclerAdapter;
```

```
import com.firebase.ui.database.FirebaseRecyclerOptions;
import com.google.android.material.floatingactionbutton.FloatingActionButton;
import com.google.android.material.navigation.NavigationView;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.squareup.picasso.Picasso;

import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;
import java.util.Objects;

import de.hdodenhof.circleimageview.CircleImageView;
import io.paperdb.Paper;

public class foodItems extends AppCompatActivity implements
NavigationView.OnNavigationItemSelectedListener {
    private static final int REQUEST_CODE_SPEECH_INPUT = 1;
    private DatabaseReference ProductsRef;
    DrawerLayout drawerLayout;
    NavigationView navigationView;
    Toolbar toolbar;
    TextView textView;
    private RecyclerView recyclerView;
    RecyclerView.LayoutManager layoutManager;

    List<String> fList =  Arrays.asList("pizza","veg roll","biryani","meals","dosa");
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_grocery);
```

```
        AppCompatDelegate.setDefaultNightMode(AppCompatDelegate.MODE_NIGHT_YES);
        ProductsRef = FirebaseDatabase.getInstance().getReference().child("Food");

       drawerLayout=findViewById(R.id.drawer_layout1);
        navigationView=findViewById(R.id.nav_view1);
        toolbar=findViewById(R.id.category_toolbar);
//        Toolbar toolbar = (Toolbar) findViewById(R.id.category_toolbar);
       toolbar.setTitle("Home");
//        setSupportActionBar(toolbar);

        navigationView.bringToFront();
        ActionBarDrawerToggle toggle=new
ActionBarDrawerToggle(this,drawerLayout,toolbar,R.string.navigation_drawer_open,R.string.na
vigation_drawer_close);
        drawerLayout.addDrawerListener(toggle);
        toggle.syncState();
        navigationView.setNavigationItemSelectedListener(this);



        View headerView = navigationView.getHeaderView(0);
        TextView userNameTextView = headerView.findViewById(R.id.user_profile_name);
        CircleImageView profileImageView = headerView.findViewById(R.id.user_profile_image);
        userNameTextView.setText(Prevalent.currentOnlineUser.getName());

Picasso.get().load(Prevalent.currentOnlineUser.getImage()).placeholder(R.drawable.profile).into
(profileImageView);
        recyclerView = findViewById(R.id.recycler_menu);
        recyclerView.setHasFixedSize(true);
        layoutManager = new LinearLayoutManager(this);
        recyclerView.setLayoutManager(layoutManager);
```

```
        FloatingActionButton fab = (FloatingActionButton) findViewById(R.id.fab1);
        fab.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent = new Intent(foodItems.this,CartActivity.class);
                startActivity(intent);
            }
        });

    }

    public void speakButtonClicked(View v) {
        startVoiceRecognitionActivity();
    }
    @Override
    protected void onStart() {
        super.onStart();
        FirebaseRecyclerOptions<Products> options =
                new FirebaseRecyclerOptions.Builder<Products>()
                        .setQuery(ProductsRef, Products.class)
                        .build();

        FirebaseRecyclerAdapter<Products, ProductViewHolder> adapter =
                new FirebaseRecyclerAdapter<Products, ProductViewHolder>(options) {
                    @Override
                    protected void onBindViewHolder(@NonNull ProductViewHolder holder, int
position, @NonNull final Products model)
                    {
                        holder.txtProductName.setText(model.getPname());
                        holder.txtProductDescription.setText(model.getDescription());
                        holder.txtProductPrice.setText("Price = " + model.getPrice() + "Rs.");
```

```
                Picasso.get().load(model.getImage()).into(holder.imageView);
                holder.itemView.setOnClickListener(new View.OnClickListener() {
                    @Override
                    public void onClick(View view) {
                        Intent intent =new Intent(foodItems.this,FoodProductDetailsActivity.class);
                        intent.putExtra("pid",model.getPid());
                        startActivity(intent);
                    }
                });
            }

            @NonNull
            @Override
            public ProductViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int
viewType) {
                View view =
LayoutInflater.from(parent.getContext()).inflate(R.layout.product_items_layout, parent, false);
                ProductViewHolder holder = new ProductViewHolder(view);
                return holder;
            }
        };
    recyclerView.setAdapter(adapter);
    adapter.startListening();

  }
  @Override
  public void onBackPressed(){
    if(drawerLayout.isDrawerOpen(GravityCompat.START)){
      drawerLayout.closeDrawer(GravityCompat.START);
    }
    else
```

```
        {super.onBackPressed();
        }
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        // Inflate the menu; this adds items to the action bar if it is present.
        getMenuInflater().inflate(R.menu.main_menu, menu);
        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        // Handle action bar item clicks here. The action bar will
        // automatically handle clicks on the Home/Up button, so long
        // as you specify a parent activity in AndroidManifest.xml.
        int id = item.getItemId();

        //noinspection SimplifiableIfStatement
//        if (id == R.id.action_settings) {
//            return true;
//        }

        return super.onOptionsItemSelected(item);
    }




    @Override
    public boolean onNavigationItemSelected(MenuItem item) {
```

```
        // Handle navigation view item clicks here.
        int id = item.getItemId();

        if (id == R.id.nav_cart) {
            Intent intent = new Intent(foodItems.this,CartActivity.class);
            startActivity(intent);
        } else if (id == R.id.nav_search) {
            Intent intent = new Intent(foodItems.this,SearchProductsActivity.class);
            startActivity(intent);

        } else if (id == R.id.nav_categories) {

        } else if (id == R.id.nav_settings) {
            Intent intent=new Intent(foodItems.this,SettinsActivity.class);
            startActivity(intent);

        } else if (id == R.id.nav_logout) {
            Paper.book().destroy();
            Intent intent=new Intent(foodItems.this,MainActivity.class);
            intent.addFlags(Intent.FLAG_ACTIVITY_NEW_TASK
|Intent.FLAG_ACTIVITY_CLEAR_TASK );
            startActivity(intent);
            finish();

        }


        drawerLayout.closeDrawer(GravityCompat.START);
        return true;
    }
```

```
private void startVoiceRecognitionActivity() {
    Intent intent = new Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH);
    intent.putExtra(RecognizerIntent.EXTRA_LANGUAGE_MODEL,
            RecognizerIntent.LANGUAGE_MODEL_FREE_FORM);
    intent.putExtra(RecognizerIntent.EXTRA_PROMPT, "Say your Product...");
    intent.putExtra(RecognizerIntent.EXTRA_MAX_RESULTS, 1);

    startActivityForResult(intent, REQUEST_CODE_SPEECH_INPUT);
}



@Override
protected void onActivityResult(int requestCode, int resultCode, Intent data) {
    String tv_Speech_to_text = "";
    String parameter = "";
    super.onActivityResult(requestCode, resultCode, data);

    if (requestCode == REQUEST_CODE_SPEECH_INPUT) {
        if (resultCode == RESULT_OK && data != null) {
            ArrayList<String> result = data.getStringArrayListExtra(
                    RecognizerIntent.EXTRA_RESULTS);
            tv_Speech_to_text = Objects.requireNonNull((result).get(0));

            if (tv_Speech_to_text.equals("cart")|| tv_Speech_to_text.equals("karte")) {
                Intent intent = new Intent(foodItems.this,CartActivity.class);
                startActivity(intent);
            }
        } else if(tv_Speech_to_text.equals("category")|| tv_Speech_to_text.equals("back")) {
            Toast.makeText(this, "back pressed", Toast.LENGTH_SHORT).show();
            super.onBackPressed();
        }
```

```
        for(String item:fList)
        {
            if(tv_Speech_to_text.equals(item)) {
                Toast.makeText(this, item, Toast.LENGTH_SHORT).show();

                if(item.equals("pizza"))
                    parameter = "Jul 31, 202200:17:50 am";
                else if(item.equals("veg roll"))
                    parameter = "Jul 31, 202200:18:19 am";
                else if(item.equals("biryani"))
                    parameter = "Jul 31, 202200:18:57 am";
                else if(item.equals("meals"))
                    parameter = "Jul 31, 202200:19:32 am";
                else if(item.equals("dosa"))
                    parameter = "Jul 31, 202200:19:58 am";
                Intent intent = new Intent(foodItems.this, FoodProductDetailsActivity.class);
                intent.putExtra("pid", parameter);
                startActivity(intent);
            }
        }

    }
  }
}
```

Cart Activity

```
package com.dhruva.shopping;

import android.content.DialogInterface;
import android.content.Intent;
```

```
import android.os.Bundle;

import android.speech.RecognizerIntent;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.ConcatAdapter;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import     androidx.recyclerview.widget.RecyclerView.Adapter;

import com.dhruva.shopping.Model.Cart;
import com.dhruva.shopping.Prevalent.CartViewHolder;
import com.dhruva.shopping.Prevalent.Prevalent;
import com.firebase.ui.database.FirebaseRecyclerAdapter;
import com.firebase.ui.database.FirebaseRecyclerOptions;
import com.google.android.gms.tasks.OnCompleteListener;
import com.google.android.gms.tasks.Task;
import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;
```

```
import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;
import java.util.Objects;

public class CartActivity<val> extends AppCompatActivity {
    private static final int REQUEST_CODE_SPEECH_INPUT = 1;
    private RecyclerView recyclerView;
    private RecyclerView recyclerView1;
    private RecyclerView.LayoutManager layoutManager, layoutManager1;
    private Button NextProcessBtn;
    private TextView txtTotalAmount, txtMsg1;
    private int overTotalPrice = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_cart);
        recyclerView = findViewById(R.id.cart_list);
        recyclerView.setHasFixedSize(true);
        layoutManager = new LinearLayoutManager(this);
        recyclerView.setLayoutManager(layoutManager);


        NextProcessBtn = (Button) findViewById(R.id.next_btn);
        txtTotalAmount = (TextView) findViewById(R.id.total_price);
        txtMsg1 = (TextView) findViewById(R.id.msg1);
        NextProcessBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
```

```
            txtTotalAmount.setText("Total Price = Rs." + String.valueOf(overTotalPrice));
            Intent intent = new Intent(CartActivity.this, ConfirmFinalOrderActivity.class);
            intent.putExtra("Total Price", String.valueOf(overTotalPrice));
            startActivity(intent);
            finish();
        }
    });

}

public void speakButtonClicked(View v) {
    startVoiceRecognitionActivity();
}

@Override
protected void onStart() {
    super.onStart();

    FirebaseRecyclerAdapter<Cart, CartViewHolder> adapter,adapter1;

    // CheckOrderState();
    List<String> cart_category = Arrays.asList("Products", "ELectronics", "Pharmacy",
"Food");

    final DatabaseReference cartListRef =
FirebaseDatabase.getInstance().getReference().child("Cart List");

    FirebaseRecyclerOptions<Cart> options =
            new FirebaseRecyclerOptions.Builder<Cart>()
                    .setQuery(cartListRef.child("User view")
```

```
                    .child(Prevalent.currentOnlineUser.getPhone()).child("Products"),
Cart.class).build();

        adapter
            = new FirebaseRecyclerAdapter<Cart, CartViewHolder>(options) {
          @Override
          protected void onBindViewHolder(@NonNull CartViewHolder holder, int position,
@NonNull final Cart model) {
            holder.txtProductQuantity.setText("Quantity = " + model.getQuantity());
            holder.txtProductPrice.setText("Price = " + model.getPrice() + " Rs.");
            holder.txtProductName.setText(model.getPname());
            int oneTyprProductTPrice = ((Integer.valueOf(model.getPrice()))) *
Integer.valueOf(model.getQuantity());
            overTotalPrice = overTotalPrice + oneTyprProductTPrice;

            holder.itemView.setOnClickListener(new View.OnClickListener() {
              @Override
              public void onClick(View view) {
                CharSequence options[] = new CharSequence[]
                        {
                            "Edit",
                            "Remove"
                        };
                AlertDialog.Builder builder = new AlertDialog.Builder(CartActivity.this);
                builder.setTitle("Cart Options: ");
                builder.setItems(options, new DialogInterface.OnClickListener() {
                  @Override
                  public void onClick(DialogInterface dialogInterface, int i) {
                    if (i == 0) {
                      Intent intent = new Intent(CartActivity.this, ProductDetailsActivity.class);
                      intent.putExtra("pid", model.getPid());
```

```
                    startActivity(intent);
                }
                if (i == 1) {
                    cartListRef.child("User view")
                            .child(Prevalent.currentOnlineUser.getPhone())
                            .child("Products")
                            .child(model.getPid())
                            .removeValue()
                            .addOnCompleteListener(new OnCompleteListener<Void>() {
                                @Override
                                public void onComplete(@NonNull Task<Void> task) {
                                    if (task.isSuccessful()) {
                                        Toast.makeText(CartActivity.this, "Item removed
Successfully.", Toast.LENGTH_SHORT).show();
                                        Intent intent = new Intent(CartActivity.this,
HomeActivity.class);
                                        startActivity(intent);
                                    }
                                }
                            });
                }
            }
        });
        builder.show();
    }
});

}

@NonNull
@Override
```

```
        public CartViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int
viewType) {
            View view =
LayoutInflater.from(parent.getContext()).inflate(R.layout.cart_items_layout, parent, false);
            CartViewHolder holder = new CartViewHolder(view);
            return holder;
        }
    };



    //Electronics Cart

    FirebaseRecyclerOptions<Cart> options1 =
            new FirebaseRecyclerOptions.Builder<Cart>()
                .setQuery(cartListRef.child("User view")
                        .child(Prevalent.currentOnlineUser.getPhone()).child("Electronics"),
Cart.class).build();

     adapter1
            = new FirebaseRecyclerAdapter<Cart, CartViewHolder>(options1) {
        @Override
        protected void onBindViewHolder(@NonNull CartViewHolder holder, int position,
@NonNull final Cart model) {
            holder.txtProductQuantity.setText("Quantity = " + model.getQuantity());
            holder.txtProductPrice.setText("Price = " + model.getPrice() + " Rs.");
            holder.txtProductName.setText(model.getPname());
            int oneTyprProductTPrice = ((Integer.valueOf(model.getPrice()))) *
Integer.valueOf(model.getQuantity());
            overTotalPrice = overTotalPrice + oneTyprProductTPrice;

            holder.itemView.setOnClickListener(new View.OnClickListener() {
```

```
@Override
public void onClick(View view) {
    CharSequence options[] = new CharSequence[]
        {
            "Edit",
            "Remove"
        };
    AlertDialog.Builder builder = new AlertDialog.Builder(CartActivity.this);
    builder.setTitle("Cart Options: ");
    builder.setItems(options, new DialogInterface.OnClickListener() {
        @Override
        public void onClick(DialogInterface dialogInterface, int i) {
            if (i == 0) {
                Intent intent = new Intent(CartActivity.this, ProductDetailsActivity.class);
                intent.putExtra("pid", model.getPid());
                startActivity(intent);
            }
            if (i == 1) {
                cartListRef.child("User view")
                        .child(Prevalent.currentOnlineUser.getPhone())
                        .child("Electronics")
                        .child(model.getPid())
                        .removeValue()
                        .addOnCompleteListener(new OnCompleteListener<Void>() {
                            @Override
                            public void onComplete(@NonNull Task<Void> task) {
                                if (task.isSuccessful()) {
                                    Toast.makeText(CartActivity.this, "Item removed
Successfully.", Toast.LENGTH_SHORT).show();
                                    Intent intent = new Intent(CartActivity.this,
HomeActivity.class);
```

```
                                    startActivity(intent);
                                }
                            }
                        });
                    }
                }
                });
                builder.show();
            }
        });

    }

    @NonNull
    @Override
    public CartViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int
viewType) {
        View view =
LayoutInflater.from(parent.getContext()).inflate(R.layout.cart_items_layout, parent, false);
        CartViewHolder holder = new CartViewHolder(view);
        return holder;
    }
  };
 //  recyclerView.setAdapter(adapter1);
  //  adapter1.startListening();

  ConcatAdapter concatenated = new ConcatAdapter(adapter,adapter1);
  recyclerView.setAdapter(adapter);
  adapter.startListening();
}
```

```
        private void buildCart(){

    }
        private void CheckOrderState() {
    /*DatabaseReference ordersRef;
    ordersRef =
FirebaseDatabase.getInstance().getReference().child("Orders").child(Prevalent.currentOnlineUse
r.getPhone());
    ordersRef.addValueEventListener(new ValueEventListener() {
        @Override
        public void onDataChange(DataSnapshot dataSnapshot) {
            if (dataSnapshot.exists()){
                String shippingState = dataSnapshot.child("state").getValue().toString();
                String userName = dataSnapshot.child("name").getValue().toString();
                if (shippingState.equals("Shipped")){
                    txtTotalAmount.setText("TDear "+userName+"\n order is shipped successfully.");
                    recyclerView.setVisibility(View.GONE);
                    txtMsg1.setVisibility(View.VISIBLE);
                    txtMsg1.setText("Congratulations, Your Final order has been shipped
successfully. Soon you will received your order at your door step.");
                    NextProcessBtn.setVisibility(View.GONE);
                    Toast.makeText(CartActivity.this,"You can purchase more products, Once you
received your first order",Toast.LENGTH_SHORT).show();
                }
                else if (shippingState.equals("Not Shipped")){
                    txtTotalAmount.setText("Shipping State = Not Shipped");
                    recyclerView.setVisibility(View.GONE);
                    txtMsg1.setVisibility(View.VISIBLE);

                    NextProcessBtn.setVisibility(View.GONE);
```

```
                Toast.makeText(CartActivity.this,"You can purchase more products, Once you
received your first order",Toast.LENGTH_SHORT).show();
            }
        }
    }

    @Override
    public void onCancelled(DatabaseError databaseError) {

    }
});*/
    }

    private void startVoiceRecognitionActivity() {
        Intent intent = new Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH);
        intent.putExtra(RecognizerIntent.EXTRA_LANGUAGE_MODEL,
                RecognizerIntent.LANGUAGE_MODEL_FREE_FORM);
        intent.putExtra(RecognizerIntent.EXTRA_PROMPT, "Say ...");
        intent.putExtra(RecognizerIntent.EXTRA_MAX_RESULTS, 1);

        startActivityForResult(intent, REQUEST_CODE_SPEECH_INPUT);
    }

    @Override
    protected void onActivityResult(int requestCode, int resultCode, Intent data) {
        String tv_Speech_to_text = "";

        super.onActivityResult(requestCode, resultCode, data);

        if (requestCode == REQUEST_CODE_SPEECH_INPUT) {
            if (resultCode == RESULT_OK && data != null) {
```

```
                ArrayList<String> result = data.getStringArrayListExtra(
                        RecognizerIntent.EXTRA_RESULTS);
                tv_Speech_to_text = Objects.requireNonNull((result).get(0));
                Toast.makeText(this, tv_Speech_to_text, Toast.LENGTH_SHORT).show();
                if (tv_Speech_to_text.equals("next")) {
                    txtTotalAmount.setText("Total Price = Rs." + String.valueOf(overTotalPrice));
                    Intent intent = new Intent(CartActivity.this, ConfirmFinalOrderActivity.class);
                    intent.putExtra("Total Price", String.valueOf(overTotalPrice));
                    startActivity(intent);
                    finish();
                }

            }
        }
    }
}
```

Confirm Final Order Activity

```
package com.dhruva.shopping;

import android.content.Intent;

import android.os.Bundle;
import android.speech.RecognizerIntent;
import android.text.TextUtils;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
```

```
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;

import com.dhruva.shopping.Prevalent.Prevalent;
import com.google.android.gms.tasks.OnCompleteListener;
import com.google.android.gms.tasks.Task;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;

import java.text.SimpleDateFormat;
import java.util.ArrayList;
import java.util.Calendar;
import java.util.HashMap;
import java.util.Objects;

public class ConfirmFinalOrderActivity extends AppCompatActivity {
    private static final int REQUEST_CODE_SPEECH_INPUT = 1;
    private EditText nameEditText, phoneEditText, addressEditText, cityEditText;
    private Button confirmOrderBtn;
    private String totalAmount = "";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_confirm_final_order);

        totalAmount = getIntent().getStringExtra("Total Price");
        Toast.makeText(this, "Total Price = Rs. " + totalAmount, Toast.LENGTH_SHORT).show();
        confirmOrderBtn = (Button) findViewById(R.id.confirm_final_order_btn);
        nameEditText = (EditText) findViewById(R.id.shippment_name);
```

```
        phoneEditText = (EditText) findViewById(R.id.shippment_phone_number);
        addressEditText = (EditText) findViewById(R.id.shippment_address);
        cityEditText = (EditText) findViewById(R.id.shippment_city);
        confirmOrderBtn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Check();
            }
        });
    }

    public void speakButtonClicked(View v) {
        startVoiceRecognitionActivity();
    }

    private void Check() {
        if (TextUtils.isEmpty(nameEditText.getText().toString())) {
            Toast.makeText(this, "Please Provide Your Full Name",
Toast.LENGTH_SHORT).show();
        } else if (TextUtils.isEmpty(phoneEditText.getText().toString())) {
            Toast.makeText(this, "Please Provide Your Phone Number",
Toast.LENGTH_SHORT).show();
        } else if (TextUtils.isEmpty(addressEditText.getText().toString())) {
            Toast.makeText(this, "Please Provide Your Valid Address.",
Toast.LENGTH_SHORT).show();
        } else if (TextUtils.isEmpty(cityEditText.getText().toString())) {
            Toast.makeText(this, "Please Provide Your City Name",
Toast.LENGTH_SHORT).show();
        } else {

            ConfirmOrder();
```

```
        }
    }

    private void ConfirmOrder() {
        final String saveCurrentTime, saveCurrentDate;
        Calendar calForDate = Calendar.getInstance();
        SimpleDateFormat currentDate = new SimpleDateFormat("MMM dd. yyy");
        saveCurrentDate = currentDate.format(calForDate.getTime());
        SimpleDateFormat currentTime = new SimpleDateFormat("HH:mm:ss a");
        saveCurrentTime = currentDate.format(calForDate.getTime());
        final DatabaseReference ordersRef = FirebaseDatabase.getInstance().getReference()
                .child("Orders")
                .child(Prevalent.currentOnlineUser.getPhone());
        HashMap<String, Object> ordersMap = new HashMap<>();
        ordersMap.put("totalAmount", totalAmount);
        ordersMap.put("name", nameEditText.getText().toString());
        ordersMap.put("phone", phoneEditText.getText().toString());
        ordersMap.put("address", addressEditText.getText().toString());
        ordersMap.put("city", cityEditText.getText().toString());
        ordersMap.put("date", saveCurrentDate);
        ordersMap.put("time", saveCurrentTime);
        ordersMap.put("state", "Not Shipped");
        ordersRef.updateChildren(ordersMap).addOnCompleteListener(new
OnCompleteListener<Void>() {
            @Override
            public void onComplete(@NonNull Task<Void> task) {
                if (task.isSuccessful()) {
                    FirebaseDatabase.getInstance().getReference()
                            .child("Cart List")
                            .child("User view")
                            .child(Prevalent.currentOnlineUser.getPhone())
```

```
                    .removeValue()
                    .addOnCompleteListener(new OnCompleteListener<Void>() {
                        @Override
                        public void onComplete(@NonNull Task<Void> task) {
                            if (task.isSuccessful()) {
                                Toast.makeText(ConfirmFinalOrderActivity.this, "Your final Order has
been placed successfully.", Toast.LENGTH_SHORT).show();
                                Intent intent = new Intent(ConfirmFinalOrderActivity.this,
HomeActivity.class);
                                intent.addFlags(Intent.FLAG_ACTIVITY_CLEAR_TASK |
Intent.FLAG_ACTIVITY_CLEAR_TASK);
                                startActivity(intent);
                                finish();
                            }
                        }
                    });
            }
        }
    });



}

    private void startVoiceRecognitionActivity() {
        Intent intent = new Intent(RecognizerIntent.ACTION_RECOGNIZE_SPEECH);
        intent.putExtra(RecognizerIntent.EXTRA_LANGUAGE_MODEL,
                RecognizerIntent.LANGUAGE_MODEL_FREE_FORM);
        intent.putExtra(RecognizerIntent.EXTRA_PROMPT, "Say ...");
        intent.putExtra(RecognizerIntent.EXTRA_MAX_RESULTS, 1);

        startActivityForResult(intent, REQUEST_CODE_SPEECH_INPUT);
```

```
    }

    @Override
    protected void onActivityResult(int requestCode, int resultCode, Intent data) {
        String tv_Speech_to_text = "";

        super.onActivityResult(requestCode, resultCode, data);

        if (requestCode == REQUEST_CODE_SPEECH_INPUT) {
            if (resultCode == RESULT_OK && data != null) {
                ArrayList<String> result = data.getStringArrayListExtra(
                        RecognizerIntent.EXTRA_RESULTS);
                tv_Speech_to_text = Objects.requireNonNull((result).get(0));
                Toast.makeText(this, tv_Speech_to_text, Toast.LENGTH_SHORT).show();
                if (tv_Speech_to_text.equals("confirm")) {
                    Check();
                }

            }
        }
    }
}
```

Output and Results

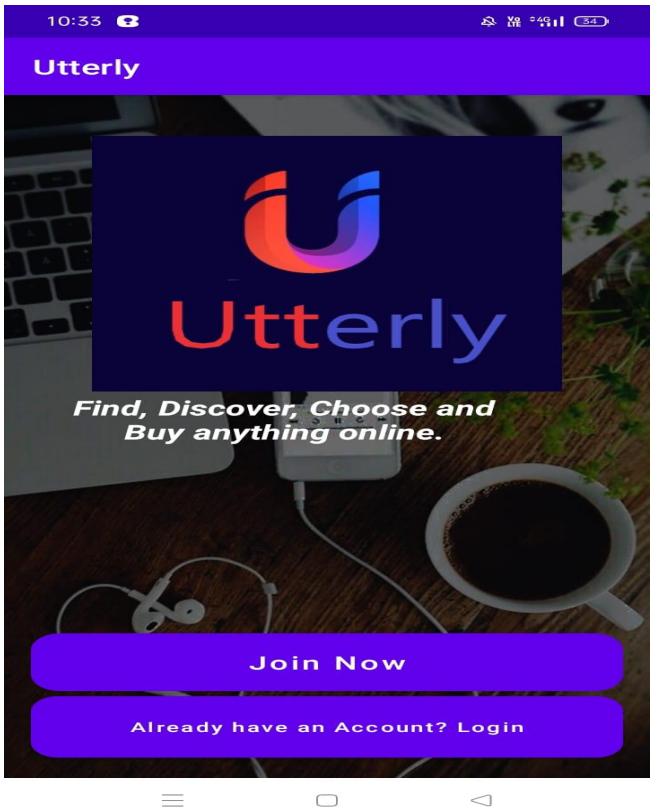

Fig 1:Main Page

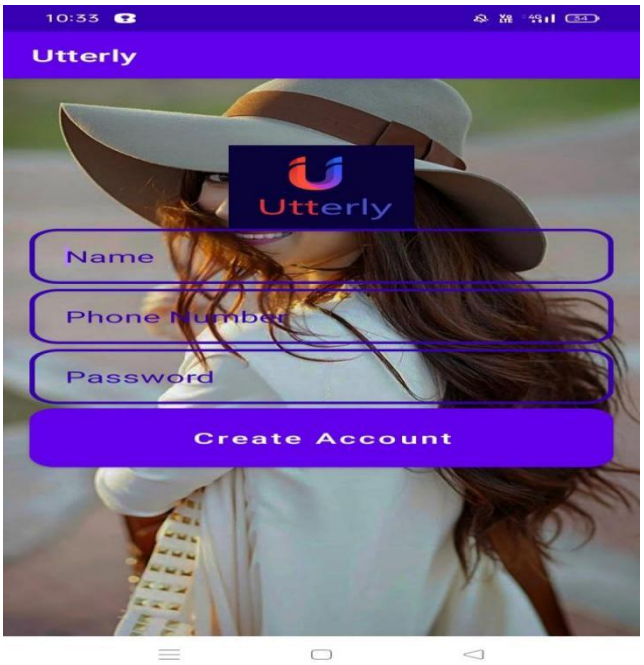

Fig 2:Register Page

Fig 3:Login Page

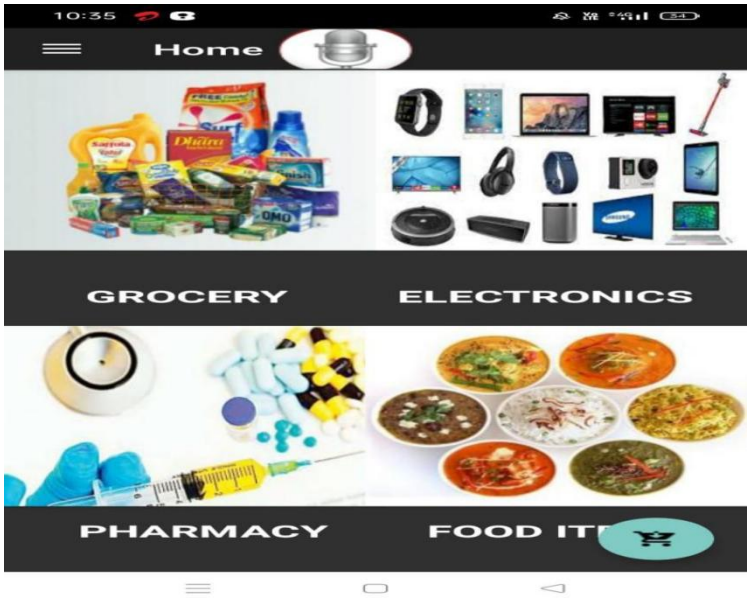

Fig 4:Home Page

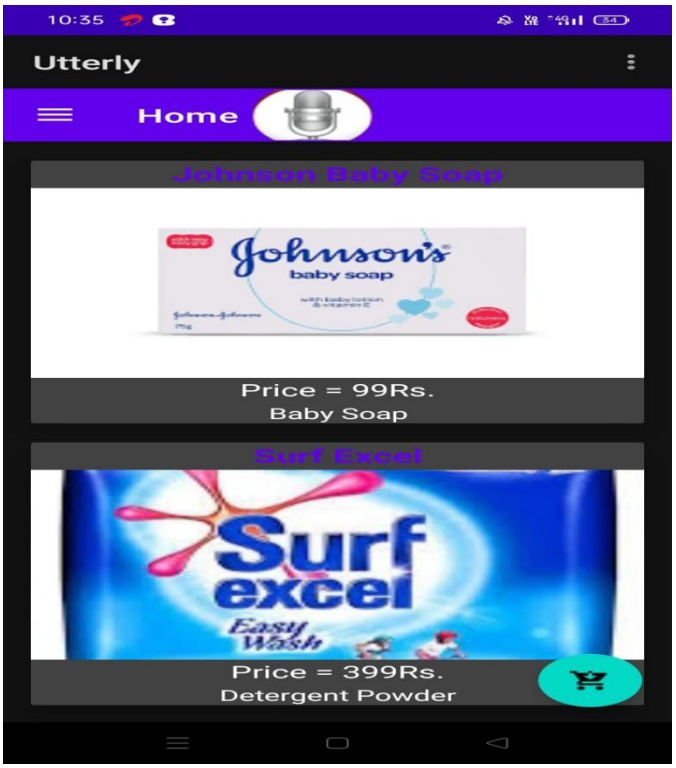

Fig 5:Category Page

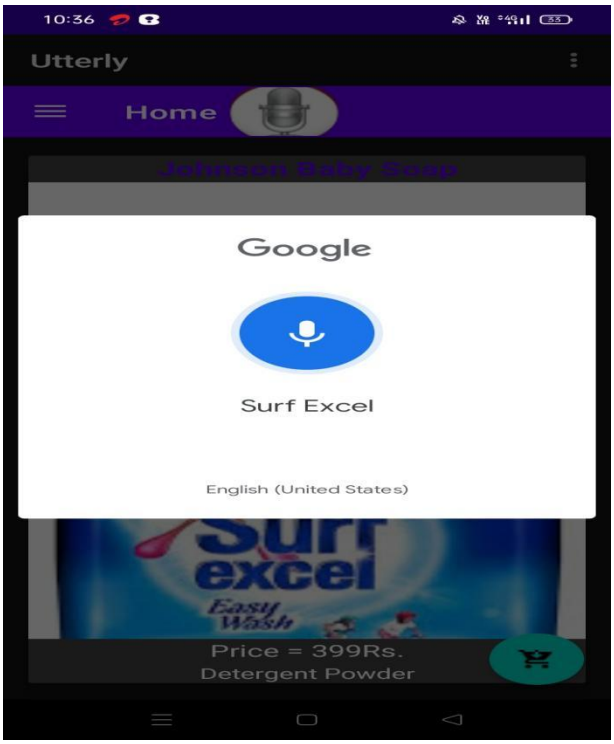

Fig 6:Voice Search Unit

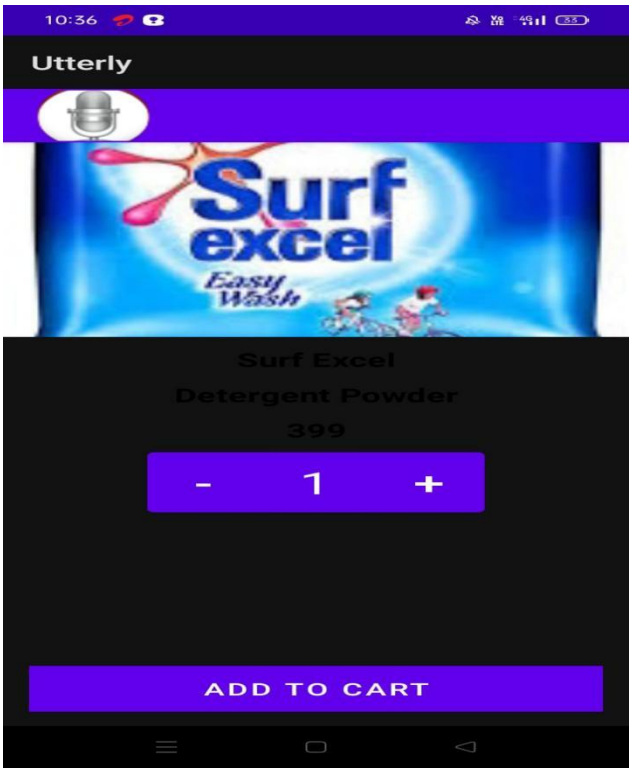

Fig 7:Item Selection Page

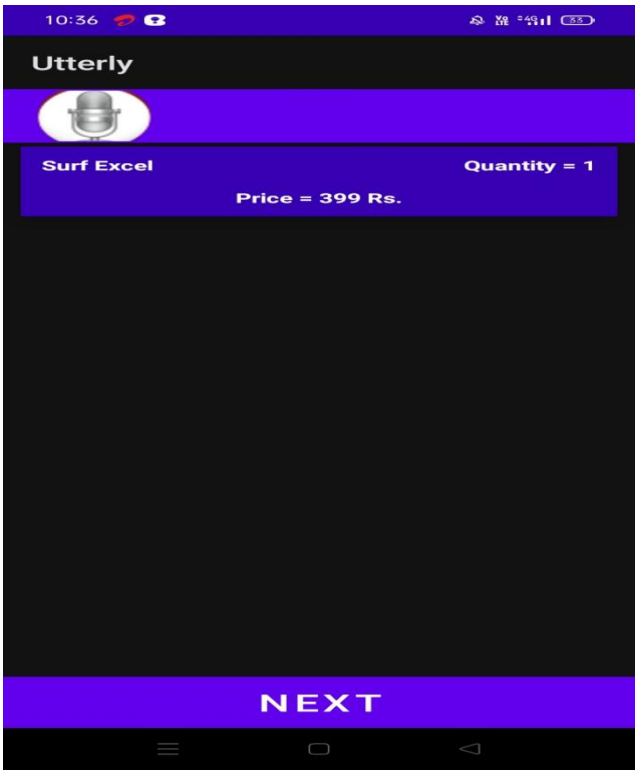

Fig 8:Cart Page

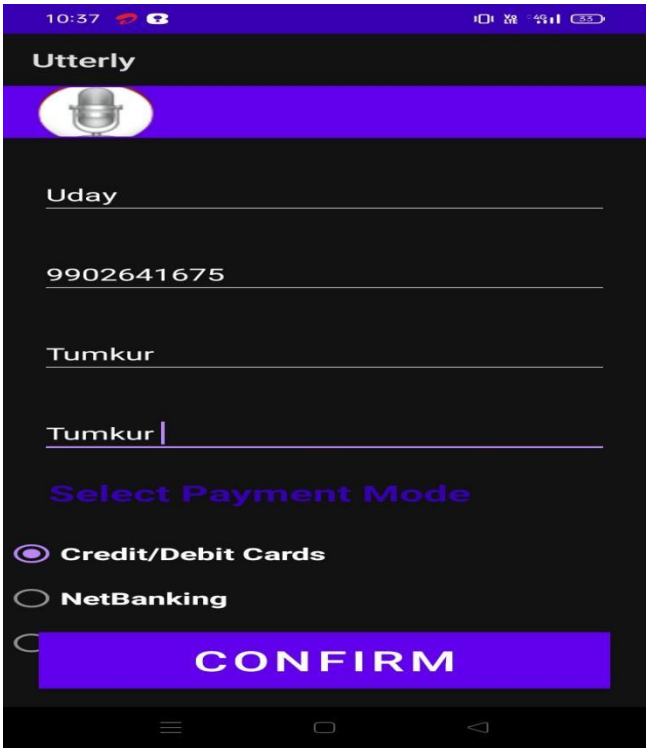

Fig 9:Address and Payment Mode

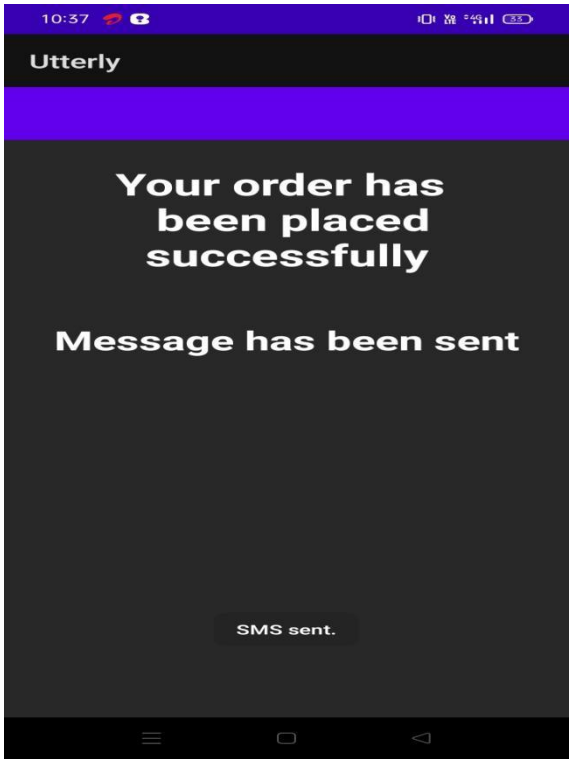

Fig 10:Order Confirmation Page

Conclusion

In this Muliple products and different cateogories can seen in on app It is user friendly app easy to understand and Mainly voice cart is added it reduces the man power The Electronic Commerce or eCommerce is a term for a business or business exchange, that includes the change of data over the internet. Online business permits clients to electronically trade products and ventures with no boundary of time or distance. The Mobile Commerce or mCommerce is the purchasing and selling of goods and services through wireless handheld gadgets, for example, mobiles and tablets. It is known as next generation eCommerce. Online shopping is a type of eCommerce which permits clients to directly purchase goods or services from a merchant over the internet using a web browser.

Traditional shopping is a tedious and time-consuming job. Although the growing trend of online shopping has reduced some load, there is still some difference in actually going to
shops and hand-picking products to get the feel of their quality and features that cannot be experienced online. Customers also feel worried to carry out online purchases due to fear of less secure transaction process that may lead to hacking of user's sensitive data, insecurity of credit/debit cards, unreliability or breach of privacy. The project aims at removing flaws of both kinds of shopping and bridge the gap between physical and a virtual world.

Biphiliography

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